

CHAPTER 3

MOTION

1.2 Motion

- 1 Define speed as distance travelled per unit time and define velocity as change in displacement per unit time
- 2 Recall and use the equation

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

$$v = \frac{s}{t}$$

- 3 Recall and use the equation

$$\text{average speed} = \frac{\text{total distance travelled}}{\text{total time taken}}$$

- 4 Define acceleration as change in velocity per unit time; recall and use the equation

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

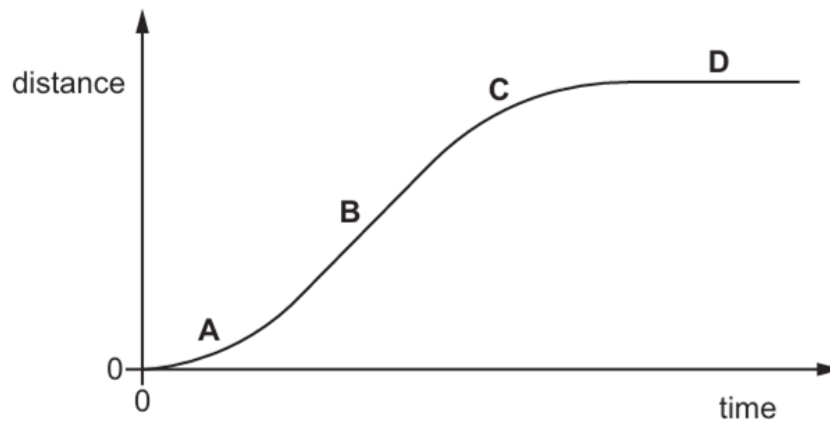
$$a = \frac{\Delta v}{\Delta t}$$

- 5 State what is meant by, and describe examples of, uniform acceleration and non-uniform acceleration
- 6 Know that a deceleration is a negative acceleration and use this in calculations
- 7 Sketch, plot and interpret distance–time and speed–time graphs
- 8 Determine from the shape of a distance–time graph when an object is:
 - (a) at rest
 - (b) moving with constant speed
 - (c) accelerating
 - (d) decelerating
- 9 Determine from the shape of a speed–time graph when an object is:
 - (a) at rest
 - (b) moving with constant speed
 - (c) moving with constant acceleration
 - (d) moving with changing acceleration
- 10 State that the acceleration of free fall g for an object near to the surface of the Earth is approximately constant and is approximately 9.8 m/s^2
- 11 Calculate speed from the gradient of a distance–time graph
- 12 Calculate the area under a speed–time graph to determine the distance travelled for motion with constant speed or constant acceleration
- 13 Calculate acceleration from the gradient of a speed–time graph

1. O/N/23/11/Q2

The diagram shows a distance–time graph for a car travelling in a straight line.

In which region is the car decelerating?



2. O/N/23/11/Q4

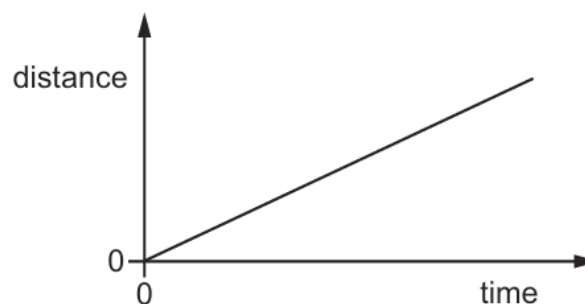
A car begins to move. It speeds up until it reaches a constant speed.

What happens to the acceleration and what happens to the velocity of the car?

- A** Both the acceleration and the velocity change.
- B** Only the acceleration changes.
- C** Only the velocity changes.
- D** Neither the acceleration nor the velocity changes.

3. O/N/23/12/Q2

The diagram shows the distance–time graph of a car.



The car is travelling along a straight road up a hill.

Which quantity for the car is constant and greater than zero?

- A** acceleration
- B** displacement
- C** gravitational potential energy
- D** kinetic energy

4. O/N/23/12/Q3

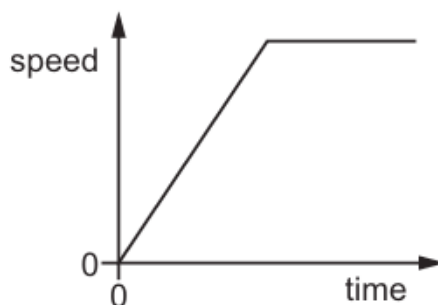
A distance–time graph and a speed–time graph show the motion of an object.

How are the speed and the acceleration of the object determined from these graphs?

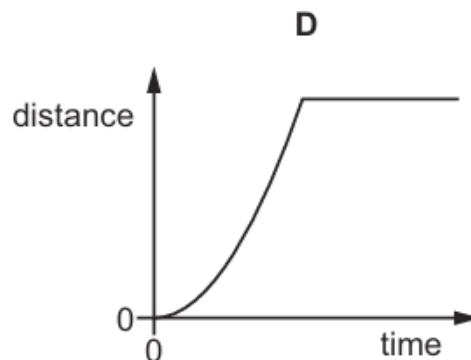
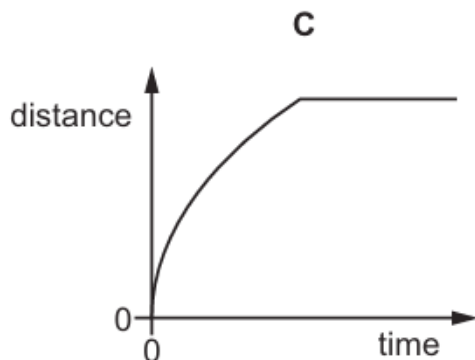
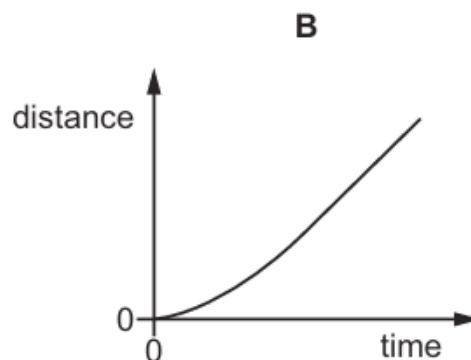
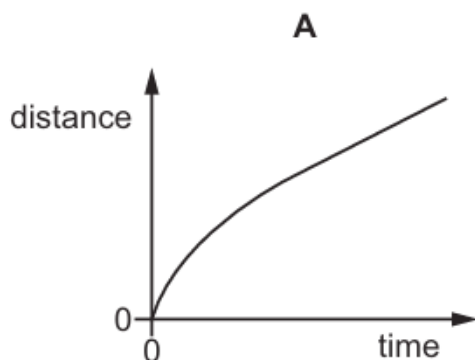
	speed	acceleration
A	area under the distance–time graph	area under the speed–time graph
B	area under the distance–time graph	gradient of the speed–time graph
C	gradient of the distance–time graph	area under the speed–time graph
D	gradient of the distance–time graph	gradient of the speed–time graph

5. M/J/23/11/Q4

A speed–time graph for the journey of a car is shown.

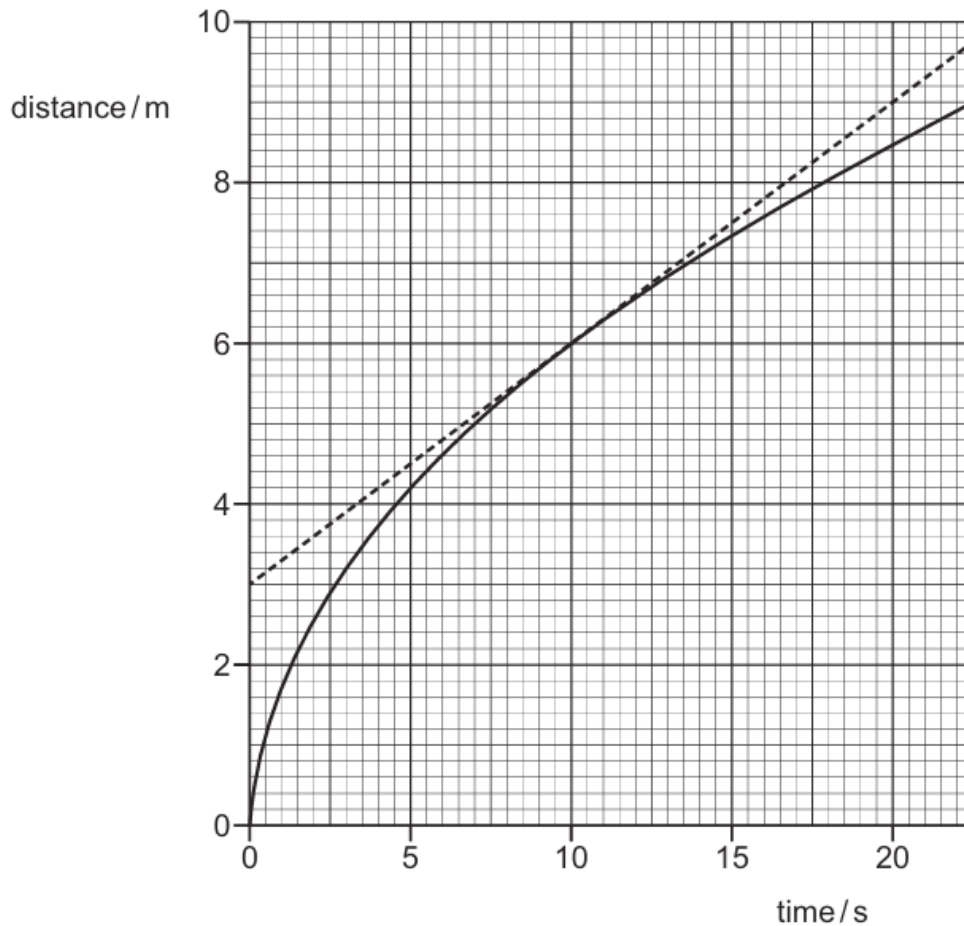


What is the distance–time graph for the journey?



6. M/J/23/12/Q4

The curved line on the distance–time graph shows the motion of a toy car. The straight line is the tangent to the curve at 10 s.



What is the speed of the toy car at 10 s?

- A** 0.3 m/s **B** 0.6 m/s **C** 1.7 m/s **D** 3.3 m/s

7. O/N/22/11/Q3

A car of mass 1000 kg is travelling down a steep hill. The brakes fail and the driver uses a horizontal sand-filled safety road to stop the car.

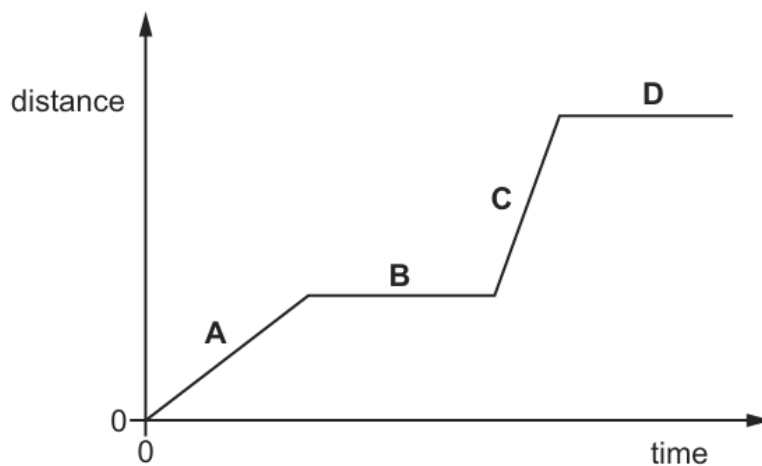
The car enters the sand at a speed of 10 m/s and experiences a constant stopping force of 2500 N.

How far does the car travel in the sand before coming to rest?

- A** 2.0 m **B** 4.0 m **C** 20 m **D** 40 m

8. O/N/22/11/Q4

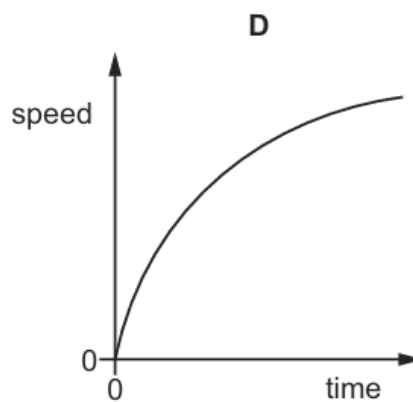
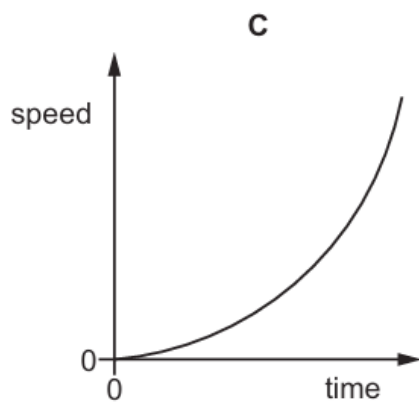
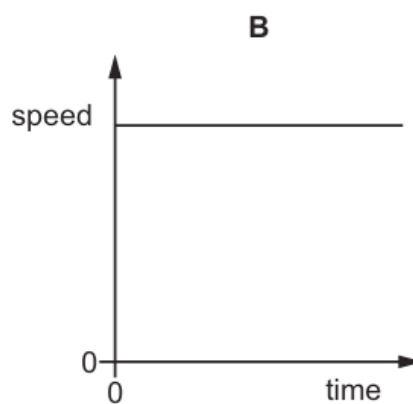
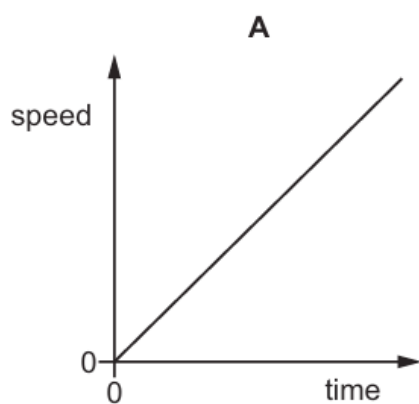
Which part of the distance–time graph shows the fastest constant speed?



9. O/N/22/11/Q5

An object is accelerating along a straight, horizontal track. The acceleration is uniform.

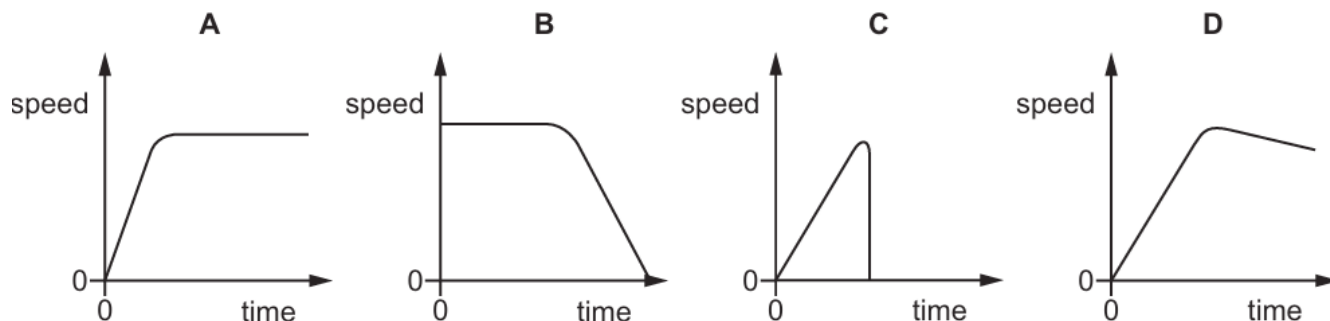
Which speed–time graph represents the motion of the object?



10. O/N/22/12/Q5

A ball starts from rest and rolls down a steep slope. The ball then rolls along rough horizontal ground.

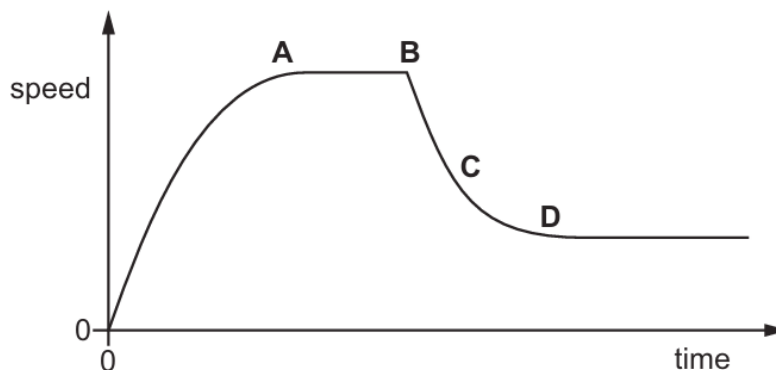
Which graph shows the speed of the ball at different times?



11. M/J/22/12/Q3

The graph shows the speed-time graph for a parachutist who jumps from a plane but does not open his parachute immediately.

At which point does he open his parachute?



12. O/N/21/11/Q3

Two identical objects begin to fall from rest.

One object falls from 200 m above the Earth's surface and the other falls from 200 m above the Moon's surface. One second after they have started to fall, both objects are still accelerating.

There is no atmosphere on the Moon and the weight of each object is constant.

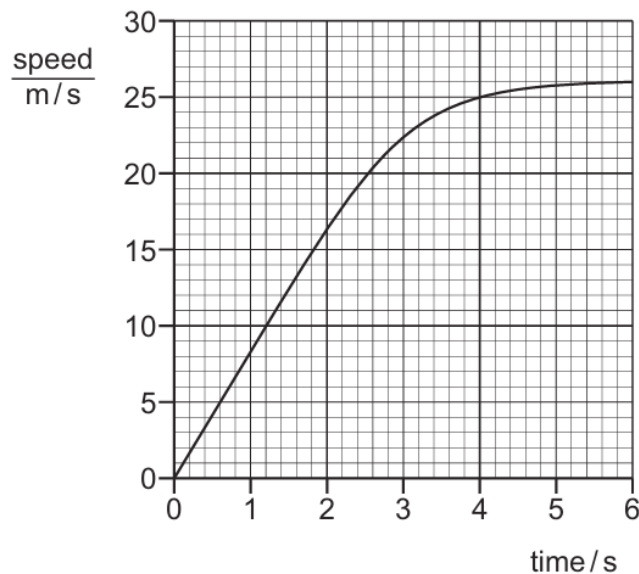
Which row describes the motion of both objects at this time?

	acceleration of object falling above the Earth's surface	acceleration of object falling above the Moon's surface
A	decreasing	constant
B	constant	constant
C	increasing	decreasing
D	increasing	increasing

13. O/N/21/12/Q3

A skier slides down a slope.

The diagram shows how his speed varies with time.

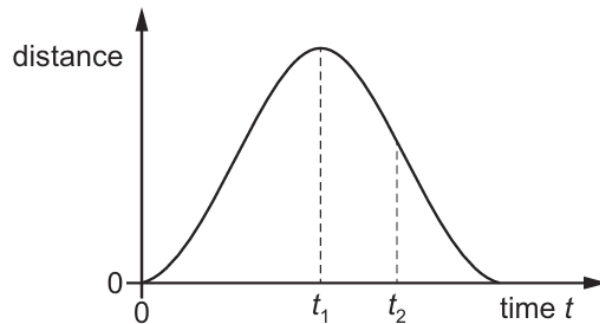


What is his average acceleration during the 6.0 s?

- A** 2.2 m/s^2 **B** 4.3 m/s^2 **C** 8.0 m/s^2 **D** 13.0 m/s^2

14. M/J/21/11/Q3

A train sets off from a station at time $t = 0$. The graph shows how the distance between the train and the station varies with time.



Which statement about the movement of the train between time t_1 and t_2 is correct?

- A** Its speed is decreasing and it is moving away from the station.
- B** Its speed is decreasing and it is moving towards the station.
- C** Its speed is increasing and it is moving away from the station.
- D** Its speed is increasing and it is moving towards the station.

15. O/N/20/11/Q2

A car begins to move. It speeds up until it reaches a constant speed. It continues to travel at this constant speed for the rest of the journey.

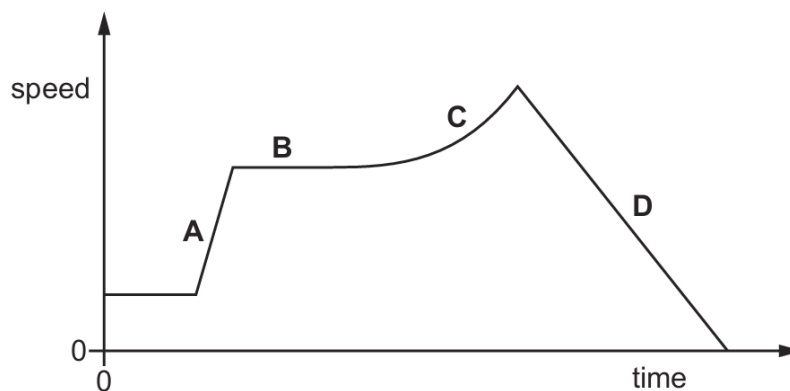
What happens to the acceleration and what happens to the velocity of the car during the journey?

- A** Both the acceleration and the velocity change.
- B** Only the acceleration changes.
- C** Only the velocity changes.
- D** Neither the acceleration nor the velocity changes.

16. O/N/20/12/Q3

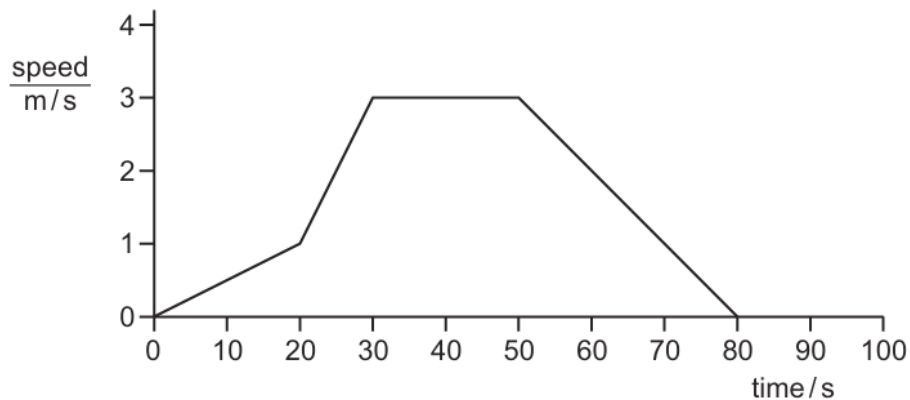
The speed–time graph for a car's journey is shown.

During which part of the journey is the car moving with non-uniform acceleration?



17. M/J/20/11/Q3

The graph shows a short journey. It is a speed–time graph.



What is the greatest speed reached?

- A** 1 m/s **B** 2 m/s **C** 3 m/s **D** 4 m/s

18. M/J/20/12/Q5

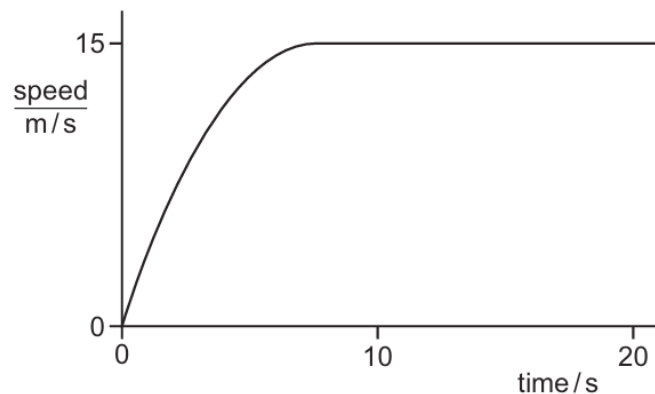
Two cameras are a known distance apart. The exact time that a vehicle passes each of the cameras is recorded.

What can be obtained from the information?

- A** average acceleration of the vehicle
B average speed of the vehicle
C maximum acceleration of the vehicle
D maximum speed of the vehicle

19. O/N/19/11/Q5

The graph shows how the speed of a car varies with time.



Which statement about the acceleration of the car between the times 10 s and 20 s is correct?

- A** The acceleration decreases.
B The acceleration increases.
C The acceleration is constant, but not zero.
D The acceleration is zero.

20. O/N/19/12/Q5

Part of a speed–time graph for two runners is shown. They are running along the same track.



Which statement **must** be correct?

- A Both runners started at the same moment.
- B Runner 1 has a greater acceleration than runner 2.
- C Runner 1 is moving faster than runner 2.
- D The distance between the two runners stays constant.

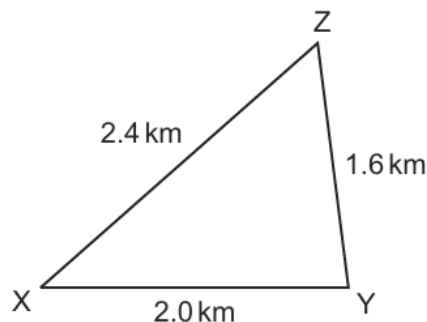
21. O/N/19/12/Q7

In which descent is the acceleration equal to the acceleration of free fall g at all times?

- A a bungee jumper leaping from a bridge
- B a feather falling in a vertical tube that contains a vacuum
- C a hailstone travelling to Earth at terminal velocity
- D a sky-diver dropping from an aircraft towards the ground

22. M/J/19/12/Q2

A woman runs 2.0 km from X to Y in 20 minutes and then rests at Y for 10 minutes. She then takes 10 minutes to run 1.6 km from Y to Z.



What is the size of her average velocity for the journey from X to Z?

- A 0.060 km/minute
- B 0.080 km/minute
- C 0.090 km/minute
- D 0.130 km/minute

23. M/J/19/12/Q3

A cyclist travelling in a straight line at 8.0 m/s accelerates to 12 m/s in a time of 6.0 s.

Which expression gives the cyclist's acceleration?

A $\frac{6}{12-8}$

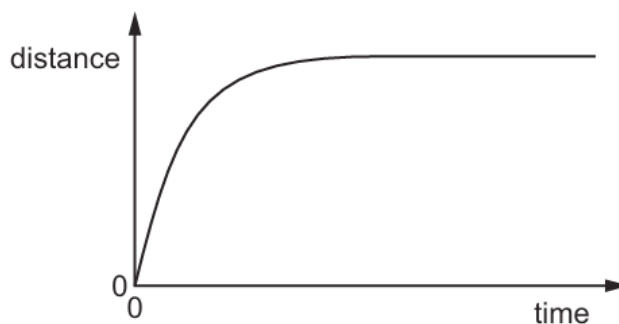
B $\frac{8}{12-6}$

C $\frac{12-6}{8}$

D $\frac{12-8}{6}$

24. M/J/19/12/Q4

The diagram shows the distance–time graph for a moving object.



What is the moving object?

A a ball thrown vertically upwards and falling back to the thrower

B a car, starting from rest, speeding up as it moves away from a traffic signal

C a rock dropped from a high cliff, falling into the sea below

D a train braking to a halt as it stops at a station

25. O/N/18/12/Q2

The table shows how the speeds of four bodies, **A**, **B**, **C** and **D**, change with time.

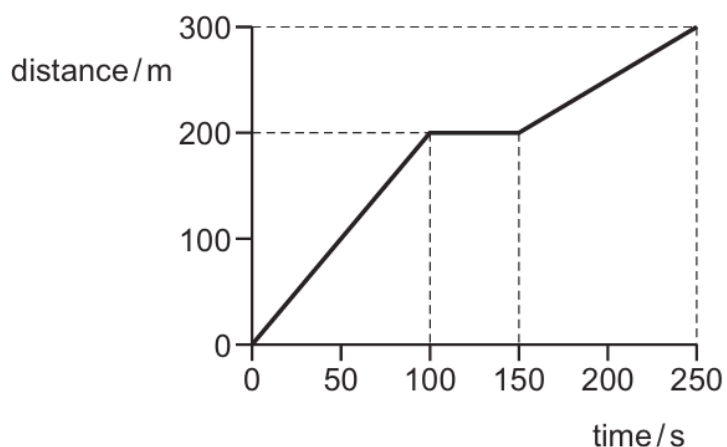
Which body has an acceleration that is **not** constant?

time/s	speed of A m/s	speed of B m/s	speed of C m/s	speed of D m/s
0	0	0	0	5.5
1	1.0	2.0	3.0	6.5
2	3.0	4.0	6.0	7.5
3	6.0	6.0	9.0	8.5

26. O/N/18/11/Q4

A cyclist takes a ride lasting 250 s.

The graph shows how the distance from the starting position varies with time.



What is his average speed for the whole journey?

- A** 1.0 m/s **B** 1.2 m/s **C** 1.5 m/s **D** 2.0 m/s

27. M/J/18/12/Q11

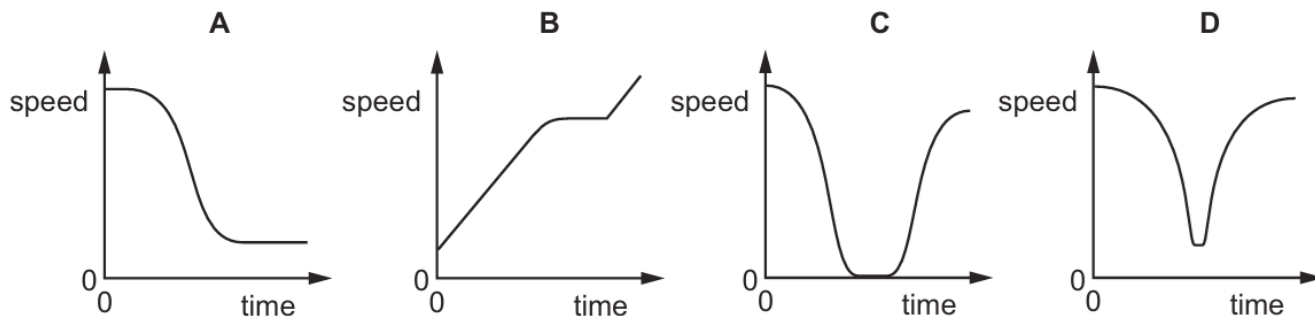
A force of 1600 N accelerates a car, of mass 800 kg, from rest.

What is the car's acceleration and its velocity after 4.0 s?

	<u>acceleration</u> m/s ²	<u>velocity</u> m/s
A	0.50	0.13
B	0.50	0.50
C	2.0	2.0
D	2.0	8.0

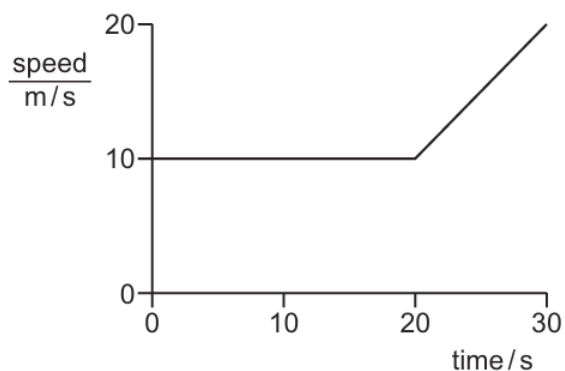
28. M/J/18/11/Q4

Which speed-time graph represents the motion of a railway train making a short stop at a station?



29. O/N/17/11/O4

An object travels for 20s with a constant speed of 10m/s. For the next 10s, it accelerates uniformly to 20m/s.

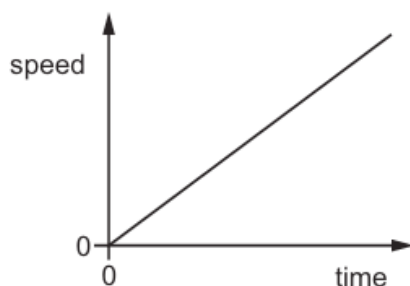


What is the total distance travelled by the object in the 30 s?

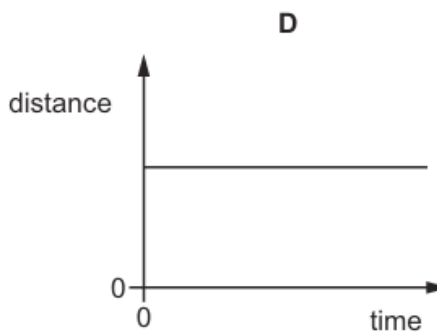
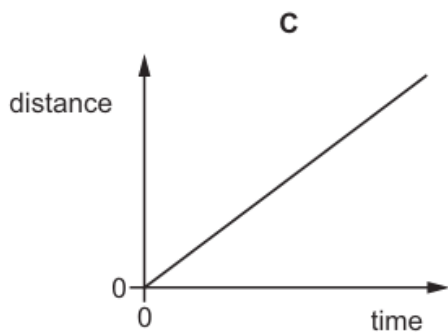
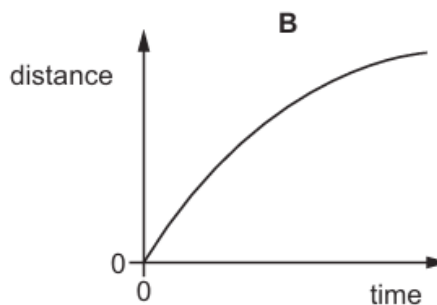
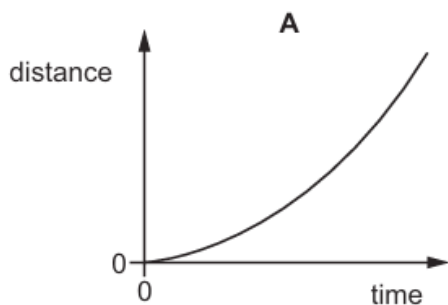
- A** 300 m **B** 350 m **C** 400 m **D** 450 m

30. O/N/17/12/Q3

The speed-time graph represents a short journey.

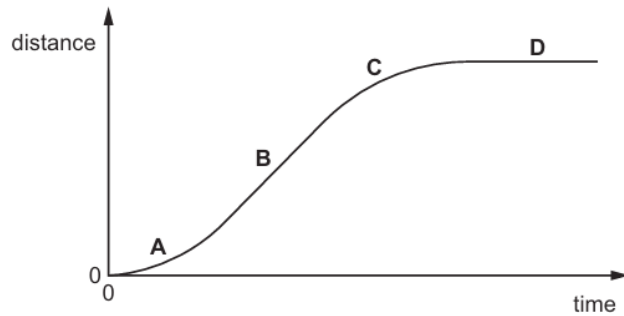


Which distance-time graph represents the same journey?



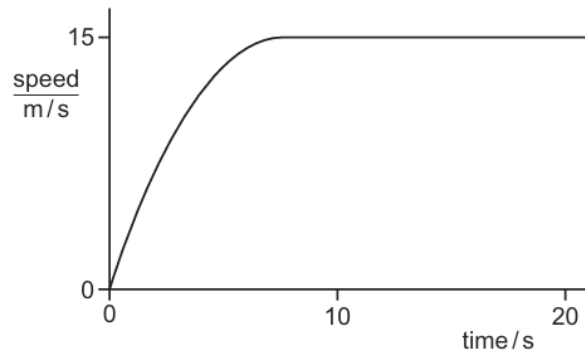
31. O/N/16/12/Q4

The graph shows a distance-time graph for a car travelling in a straight line.
In which region is the car decelerating?



32. M/J/16/11/Q4

The graph shows how the speed of a car varies with time.



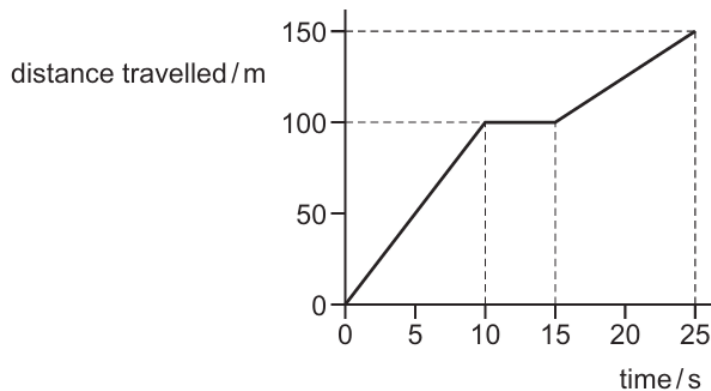
Which statement about the acceleration of the car between 10 s and 20 s is correct?

- | | |
|--------------------------------------|--|
| A The acceleration decreases. | C The acceleration is constant, but not zero. |
| B The acceleration increases. | D The acceleration is zero. |

33. M/J/15/12/Q3

A cyclist takes a ride lasting 25 s.

The diagram shows how her distance travelled from the starting position varies with time.

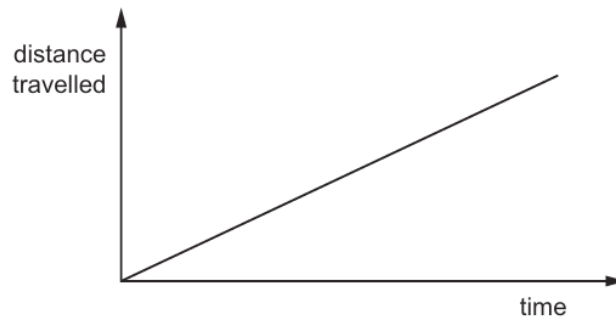


What is her average speed for the whole ride?

- | | | | |
|------------------|------------------|-------------------|-------------------|
| A 6.0 m/s | B 7.5 m/s | C 10.0 m/s | D 11.0 m/s |
|------------------|------------------|-------------------|-------------------|

34. M/J/14/12/Q6

The distance travelled by a car is increasing uniformly as it is driven along a straight road up a hill.



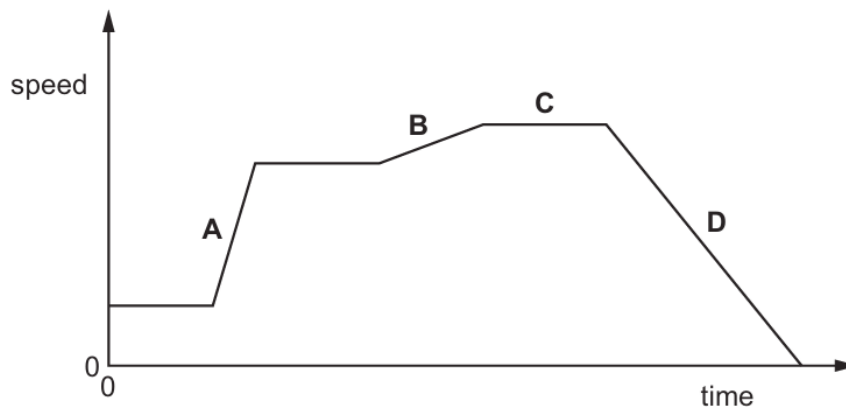
Which quantity for the car is constant but not zero?

- | | |
|---|--------------------------|
| A acceleration | C kinetic energy |
| B gravitational potential energy | D resultant force |

35. M/J/14/11/Q3

The graph shows how the speed of a car travelling in a straight line changes with time.

Which section shows the largest acceleration?



36. O/N/13/12/Q2

An object moves from P to Q in 10 s with uniform acceleration.

velocity at P = 5 m/s

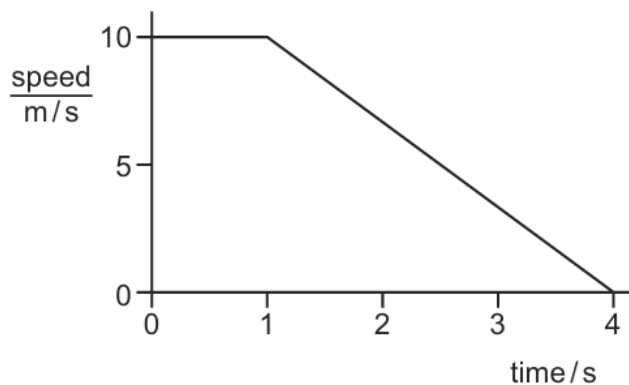
velocity at Q = 12 m/s

What is the acceleration?

- | | | | |
|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
| A 0.5 m/s ² | B 0.7 m/s ² | C 1.2 m/s ² | D 1.7 m/s ² |
|-------------------------------|-------------------------------|-------------------------------|-------------------------------|

37. M/J/13/12/Q4

The diagram shows the speed-time graph of the motion of a car for four seconds.

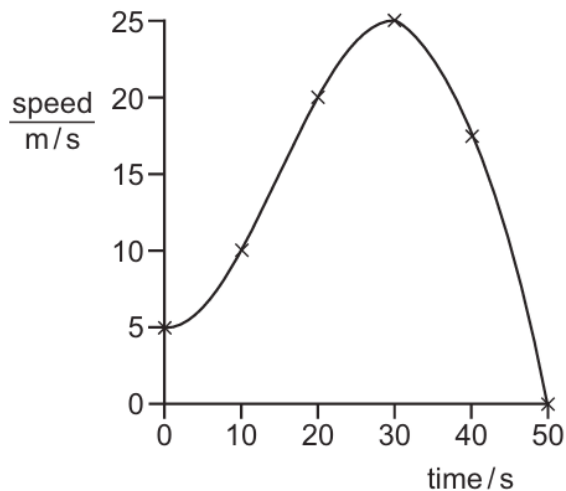


What is the distance travelled by the car in the four seconds?

- A** 15 m **B** 25 m **C** 30 m **D** 40 m

38. O/N/12/12/Q4

The speed-time graph for a car is shown.

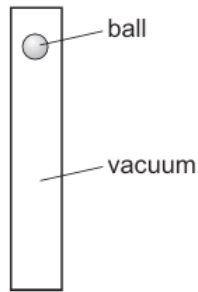


What is the acceleration of the car at 30 s?

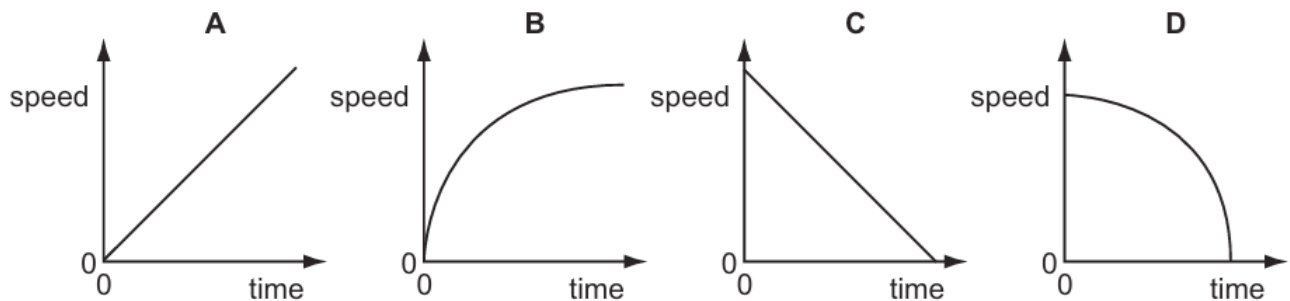
- A** 0 **B** $\frac{25-5}{30} \text{ m/s}^2$ **C** $\frac{25}{30} \text{ m/s}^2$ **D** $\frac{25}{50} \text{ m/s}^2$

39. O/N/12/12/Q3

A table-tennis ball is released from the top of an evacuated tube.



Which graph shows how the speed of the ball changes with time as it falls to the bottom of the tube?



40. O/N/12/11/Q4

A car is stationary at a set of traffic lights.

When the lights turn green, the car begins to move and continues to speed up until it reaches the maximum speed allowed. It continues to travel at this constant speed for the rest of the journey.

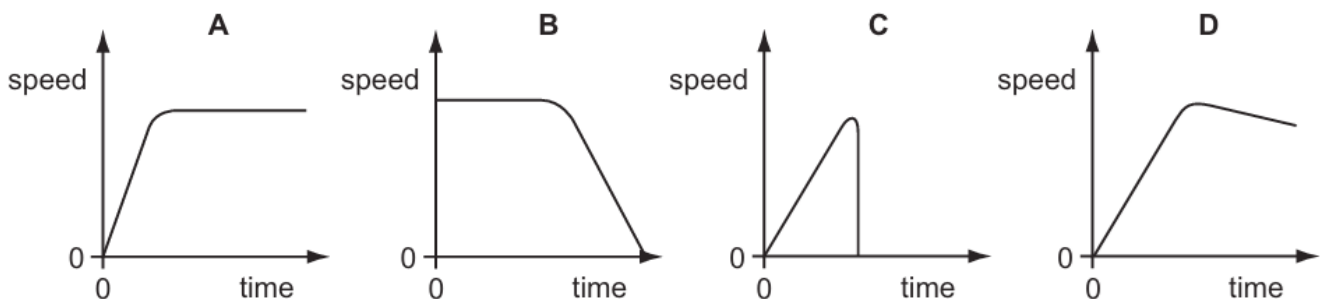
What happens to the acceleration and to the velocity of the car during this journey?

- A** Both the acceleration and the velocity change.
- B** Only the acceleration changes.
- C** Only the velocity changes.
- D** Neither the acceleration nor the velocity changes.

41. O/N/12/11/Q3

A ball starts to roll down a steep slope and then along rough horizontal ground.

Which graph best shows the **speed** of the ball?



42. M/J/12/12/Q4

A student drops a table-tennis ball in air.

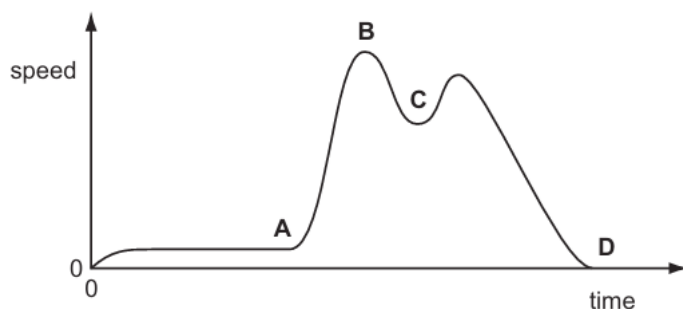
What happens to the velocity and to the acceleration of the ball during the first few seconds after release?

	velocity	acceleration
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

43. M/J/12/12/Q3

A cyclist travels along a hilly road without using the pedals or brakes. Air resistance and friction are negligible. The speed/time graph of the cyclist is shown.

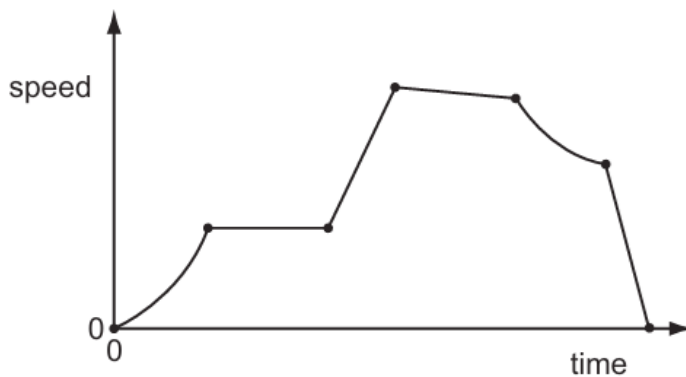
At which point did he reach the bottom of the first hill?



44. M/J/12/11/Q3

The speed-time graph represents the journey of a car.

The dots separate different sections of the journey. There are six different sections.



How many sections represent the car moving with non-uniform acceleration?

A 0

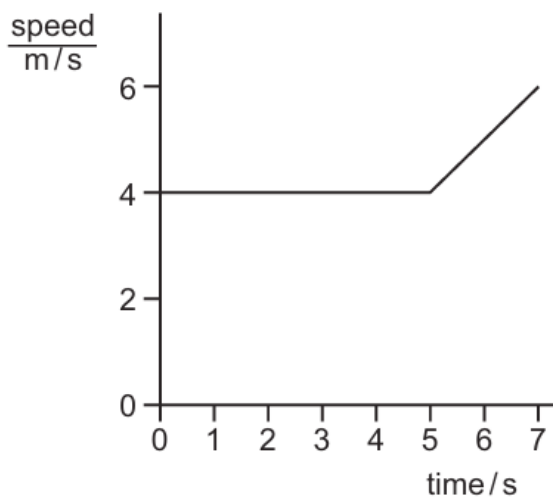
B 1

C 2

D 3

45. O/N/11/12/Q3

The graph shows part of a journey made by a cyclist.



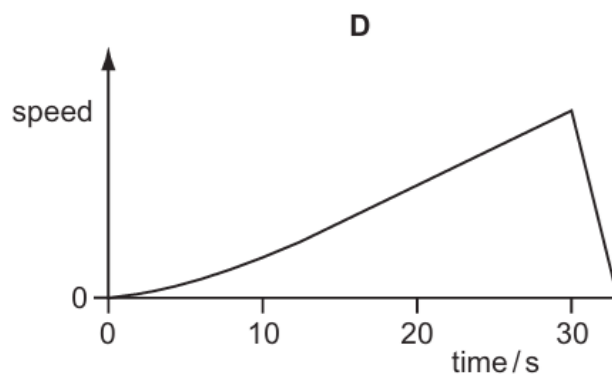
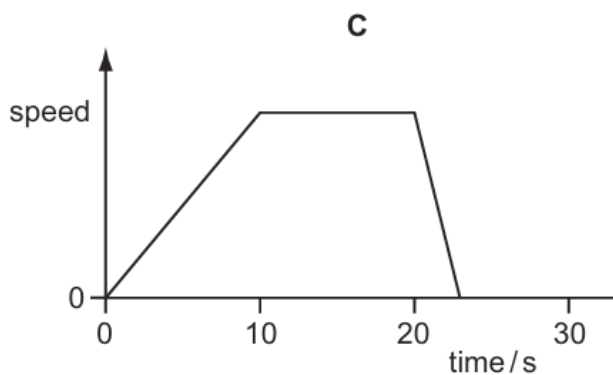
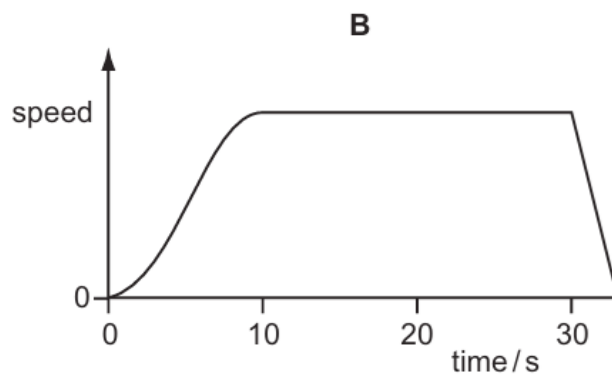
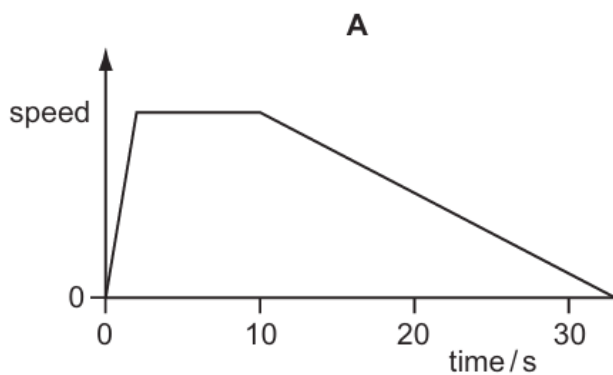
How far did the cyclist travel in 7 s?

- A** 28 m **B** 30 m **C** 32 m **D** 42 m

46. M/J/11/12/Q5

A car accelerates from traffic lights for 10 s. It stays at a steady speed for 20 s and then brakes to a stop in 3 s.

Which graph shows the journey?



47. M/J/11/12/Q4

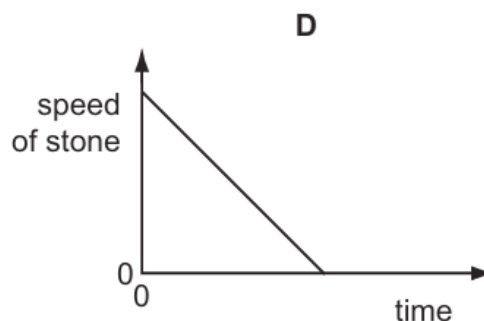
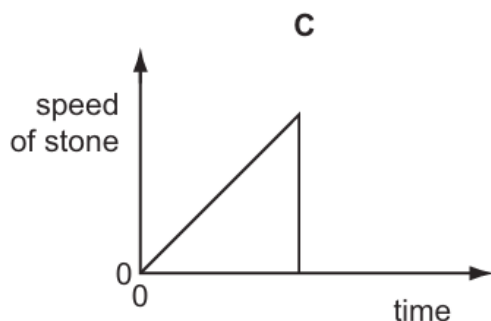
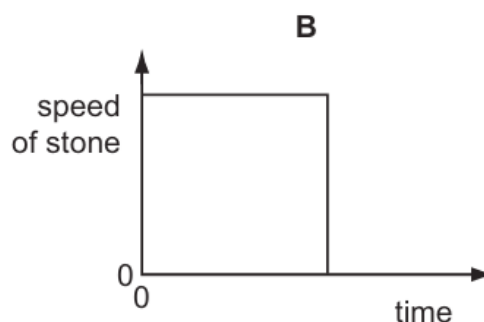
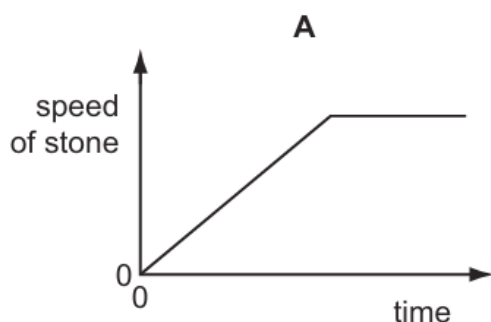
Which vehicle has an acceleration of 5 m/s^2 ?

- A** a bicycle, when its speed changes from rest to 2.5 m/s in 2 s
- B** a car, when its speed changes from rest to 15 m/s in 5 s
- C** a lorry, when its speed changes from rest to 20 m/s in 15 s
- D** a motorbike, when its speed changes from rest to 50 m/s in 10 s

48. O/N/10/12/Q2

A small stone is dropped from the top of a ladder, falls and hits the ground. It does not rebound.

Which speed-time graph is correct?



49. M/J/10/12/Q4

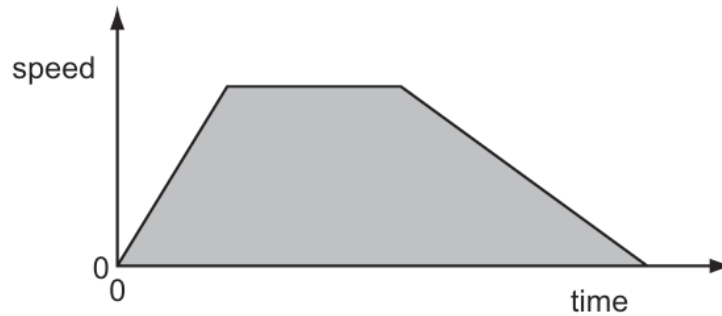
A student measures the speed of a trolley. At one instant, the speed of the trolley is 1.0 m/s and two seconds later the speed is 4.0 m/s .

What is the acceleration of the trolley?

- A** 1.5 m/s^2 **B** 2.0 m/s^2 **C** 2.5 m/s^2 **D** 5.0 m/s^2

50. M/J/10/12/Q5

The speed-time graph shows the movement of a car.



What does the shaded area of the graph represent?

- | | |
|--|--|
| A the average acceleration of the car | C the total distance travelled by the car |
| B the average speed of the car | D the total travelling time of the car |

51. M/J/10/12/Q4

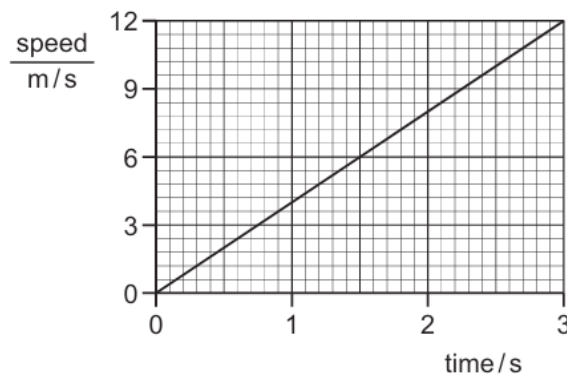
A student measures the speed of a trolley. At one instant, the speed of the trolley is 1.0 m/s and two seconds later the speed is 4.0 m/s .

What is the acceleration of the trolley?

- A** 1.5 m/s^2 **B** 2.0 m/s^2 **C** 2.5 m/s^2 **D** 5.0 m/s^2

52. O/N/09/Q4

The graph shows the speed of a car as it moves from rest.

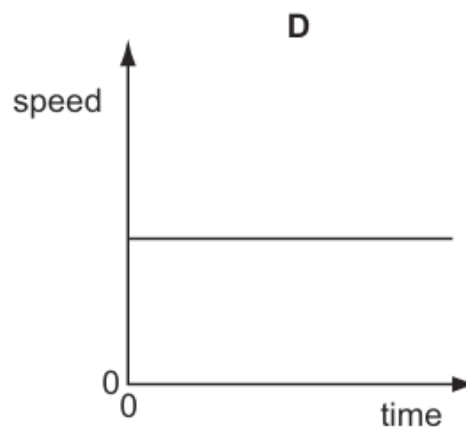
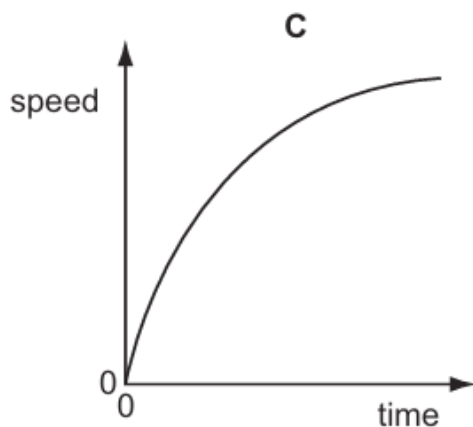
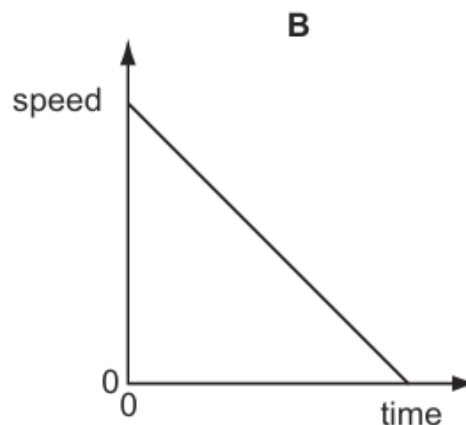
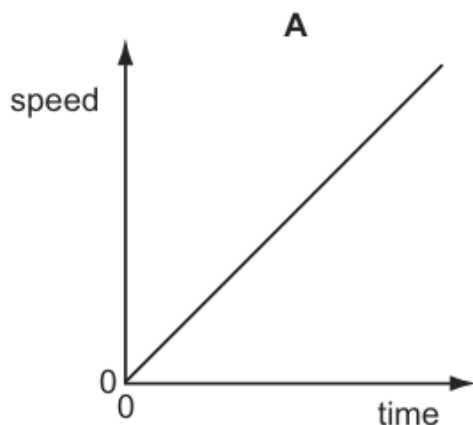


What is the average speed of the car during the first 3 s?

- A** 4 m/s **B** 6 m/s **C** 18 m/s **D** 36 m/s

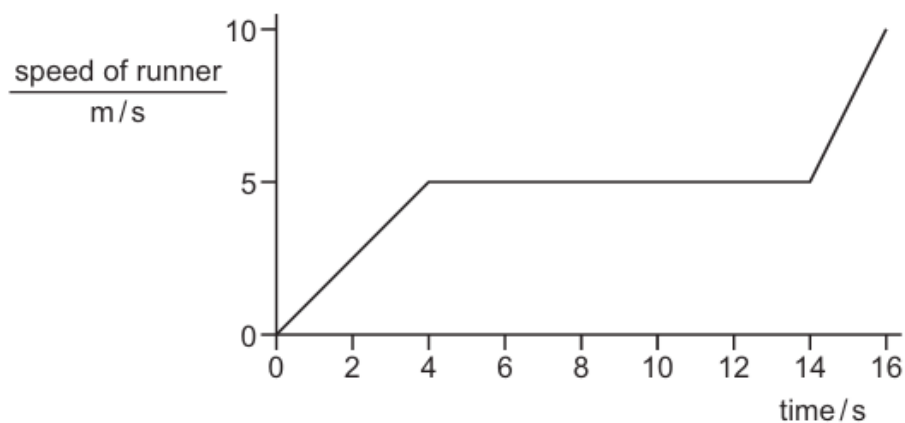
53. O/N/09/Q3

Which speed-time graph shows an object moving with non-uniform acceleration?



54. M/J/09/Q4

The graph shows the speed of a runner during a race.



What is the distance travelled by the runner during the race?

- A** 50 m **B** 65 m **C** 75 m **D** 90 m

55. O/N/08/Q3

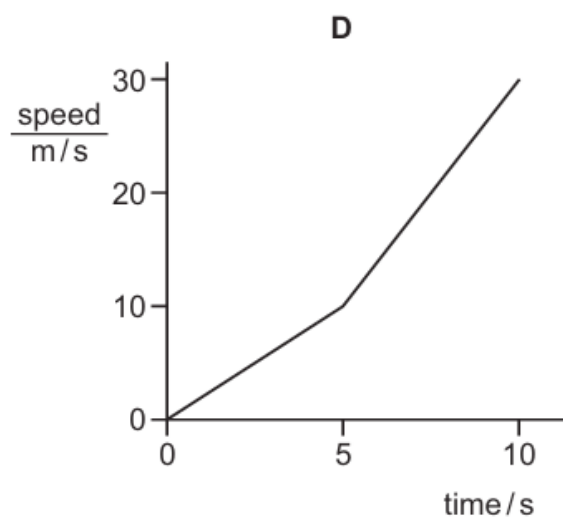
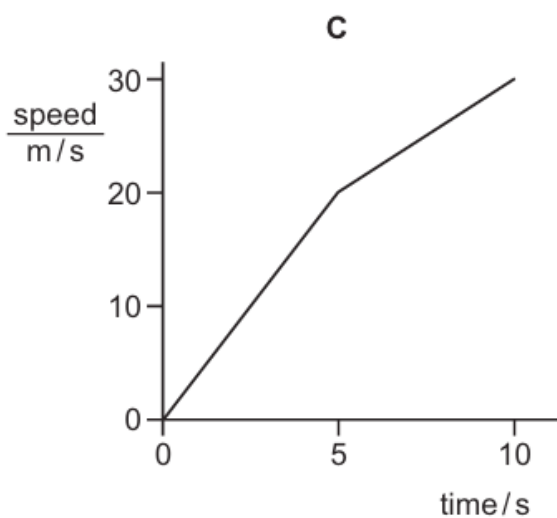
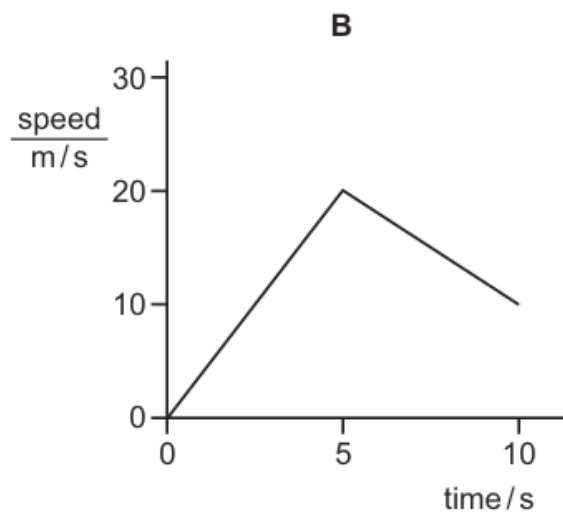
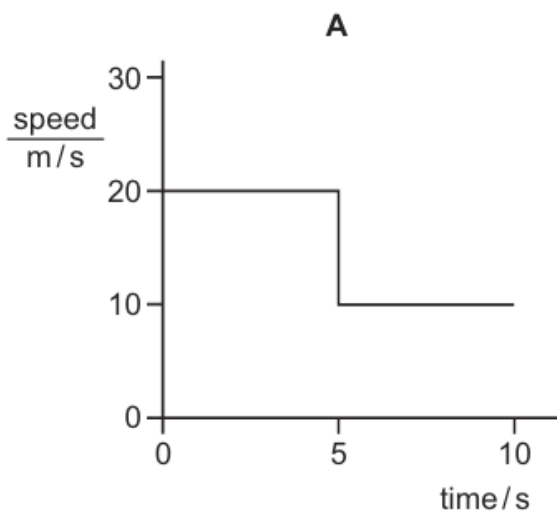
How is the motion of a body affected by balanced and unbalanced forces acting on it?

	balanced forces	unbalanced forces
A	velocity changes	velocity changes
B	velocity changes	velocity constant
C	velocity constant	velocity changes
D	velocity constant	velocity constant

56. M/J/08/Q4

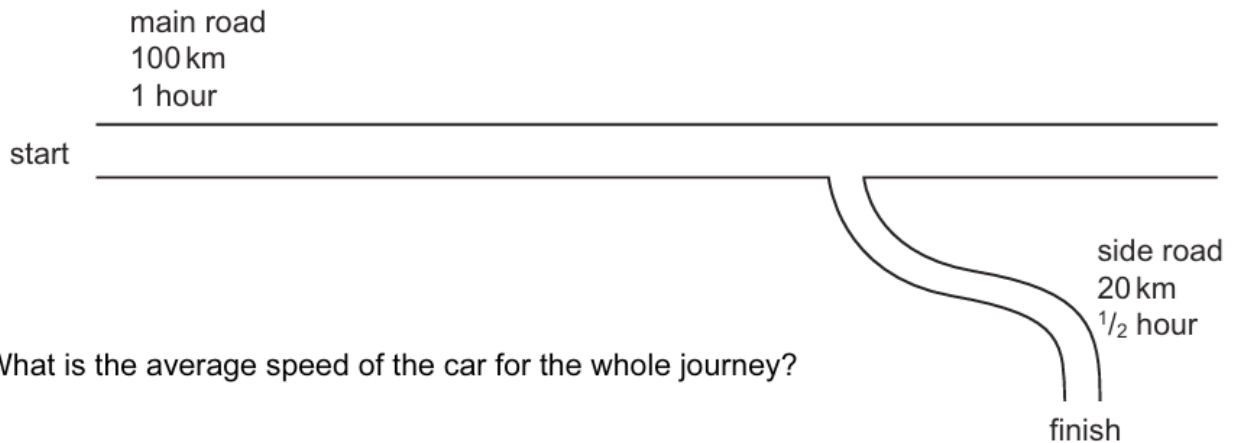
A skier is travelling downhill. The acceleration on hard snow is 4 m/s^2 and on soft snow is 2 m/s^2

Which graph shows the motion of the skier when moving from hard snow to soft snow?



57. O/N/08/Q2

A car takes 1 hour to travel 100 km along a main road and then $\frac{1}{2}$ hour to travel 20 km along a side road.

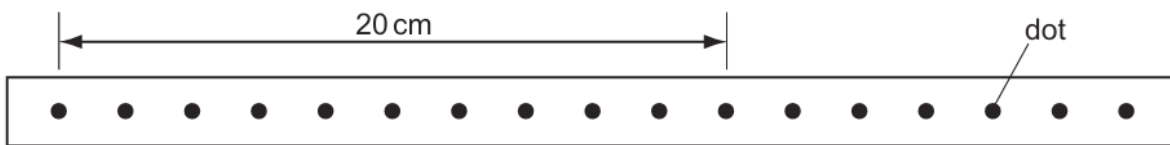


What is the average speed of the car for the whole journey?

- A** 60 km/h **C** 80 km/h
B 70 km/h **D** 100 km/h

58. O/N/06/Q5

The diagram shows a strip of paper tape that has been pulled under a vibrating arm by an object moving at constant speed. The arm was vibrating regularly, making 50 dots per second.

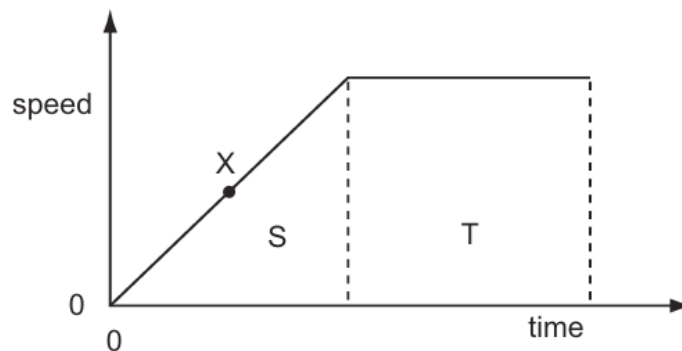


What was the speed of the object?

- A** 2.0 cm/s **B** 5.0 cm/s **C** 100 cm/s **D** 200 cm/s

59. O/N/05/Q4

The graph illustrates the motion of an object.



Which feature of the graph represents the distance travelled by the object whilst moving at a constant speed?

- A** area S **C** area T
B area S + area T **D** the gradient at point X

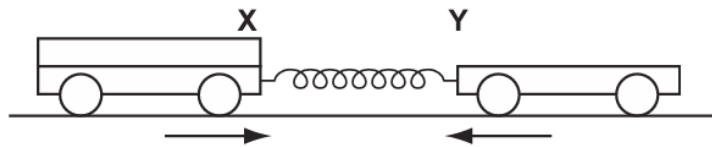
60. O/N/05/Q3

What **must** change when a body is accelerating?

- | | |
|---------------------------------------|-----------------------------------|
| A the force acting on the body | C the speed of the body |
| B the mass of the body | D the velocity of the body |

61. O/N/04/Q4

Trolley **X** and trolley **Y** are joined by a stretched spring. Trolley **X** has twice the mass of trolley **Y**.



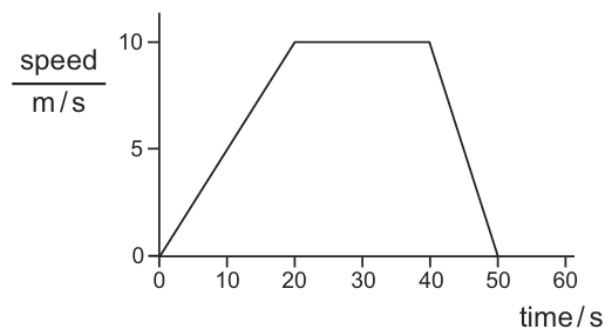
When the trolleys are released, the acceleration of **X** is 2 m/s^2 to the right.

What is the initial acceleration of trolley **Y** to the left?

- | | | | |
|----------------------------|----------------------------|----------------------------|----------------------------|
| A 1 m/s^2 | B 2 m/s^2 | C 3 m/s^2 | D 4 m/s^2 |
|----------------------------|----------------------------|----------------------------|----------------------------|

62. M/J/04/Q3

The graph shows the movement of a car over a period of 50 s.



What was the distance travelled by the car while its speed was increasing?

- | | | | |
|---------------|---------------|----------------|----------------|
| A 10 m | B 20 m | C 100 m | D 200 m |
|---------------|---------------|----------------|----------------|

63. M/J/04/Q2

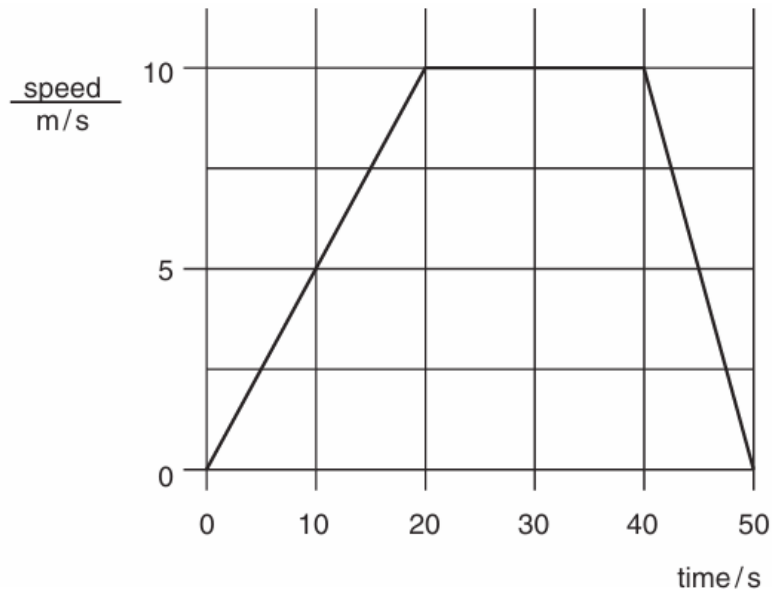
Which quantity **X** is calculated using this equation?

$$X = \frac{\text{change in velocity}}{\text{time taken}}$$

- | | |
|---------------------------|-----------------------------|
| A acceleration | C distance travelled |
| B average velocity | D speed |

64. O/N/03/Q3

The graph shows the movement of a car over a period of 50 s.



What was the distance travelled by the car during the time when it was moving at a steady speed?

- A 10 m C 200 m
B 100 m D 400 m

65. O/N/03/Q2

Which of the following **defines** acceleration?

- A $\frac{\text{change in velocity}}{\text{time taken}}$ C $\frac{\text{change in distance}}{\text{time taken}}$
B $\frac{\text{change in speed}}{\text{time taken}}$ D $\frac{\text{change in distance in a fixed direction}}{\text{time taken}}$

66. O/N/02/Q3

A car driver takes a total of two hours to make a journey of 75 km. She has a coffee break of half an hour and spends a quarter of an hour stationary in a traffic jam.

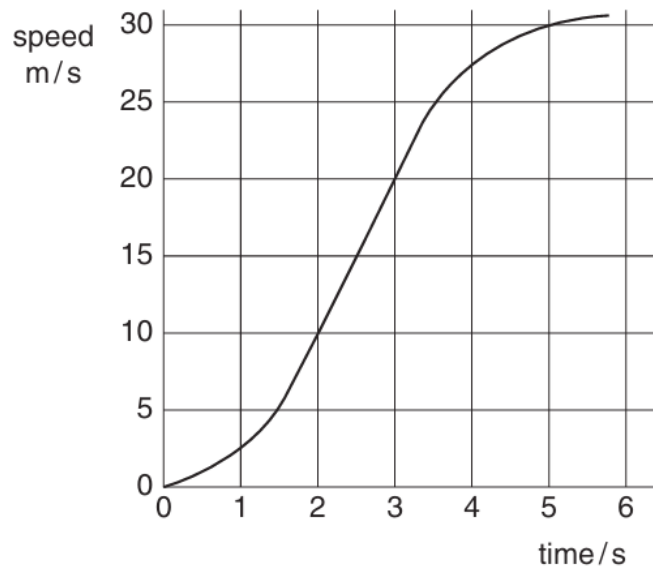
At what average speed must she travel during the rest of the time if she wants to complete the journey in the two hours?

- A 38 km/h B 50 km/h C 60 km/h D 75 km/h

67. M/J/02/Q3

The graph shows the speed of a car as it accelerates from rest.

During part of this time the acceleration is uniform.



What is the size of this uniform acceleration?

- A** 5 m/s^2 **B** 6 m/s^2 **C** 10 m/s^2 **D** 20 m/s^2

ANSWERS

- | | |
|-------|-------|
| 1. C | 48. C |
| 2. A | 49. A |
| 3. D | 50. C |
| 4. D | 51. A |
| 5. B | 52. B |
| 6. A | 53. C |
| 7. C | 54. C |
| 8. C | 55. C |
| 9. A | 56. C |
| 10. D | 57. C |
| 11. B | 58. C |
| 12. A | 59. C |
| 13. B | 60. D |
| 14. D | 61. D |
| 15. A | 62. C |
| 16. C | 63. A |
| 17. C | 64. C |
| 18. B | 65. A |
| 19. D | 66. C |
| 20. C | 67. C |
| 21. B | |
| 22. A | |
| 23. D | |
| 24. D | |
| 25. A | |
| 26. B | |
| 27. D | |
| 28. C | |
| 29. B | |
| 30. A | |
| 31. C | |
| 32. D | |
| 33. A | |
| 34. C | |
| 35. A | |
| 36. B | |
| 37. B | |
| 38. A | |
| 39. A | |
| 40. A | |
| 41. D | |
| 42. C | |
| 43. B | |
| 44. C | |
| 45. B | |
| 46. B | |
| 47. D | |

1. M/J/23/21/Q1

An aircraft pulls a glider along a runway as shown in Fig. 1.1.

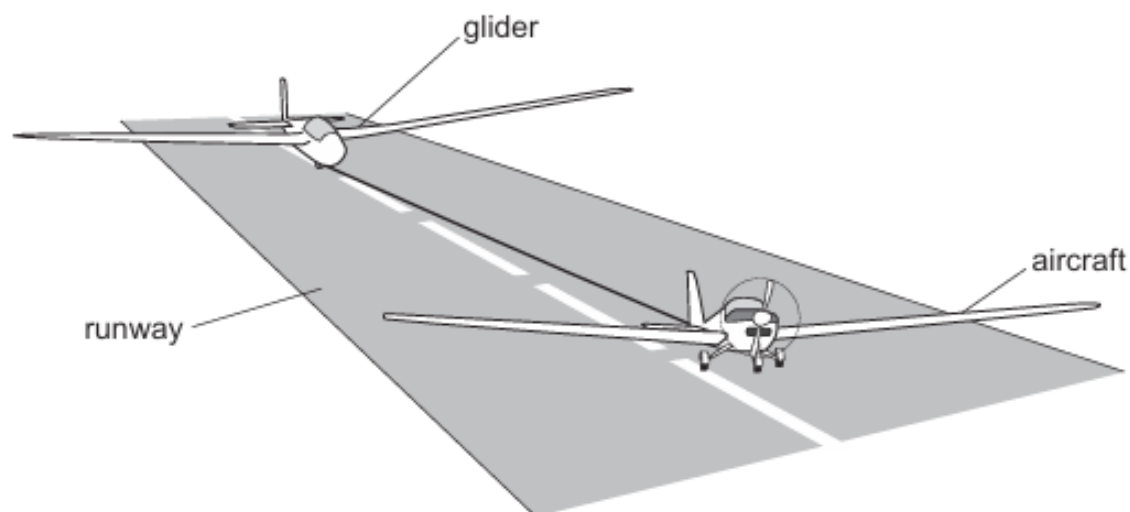


Fig. 1.1

Fig. 1.2 shows the speed of the glider during the first 12 s of the motion.

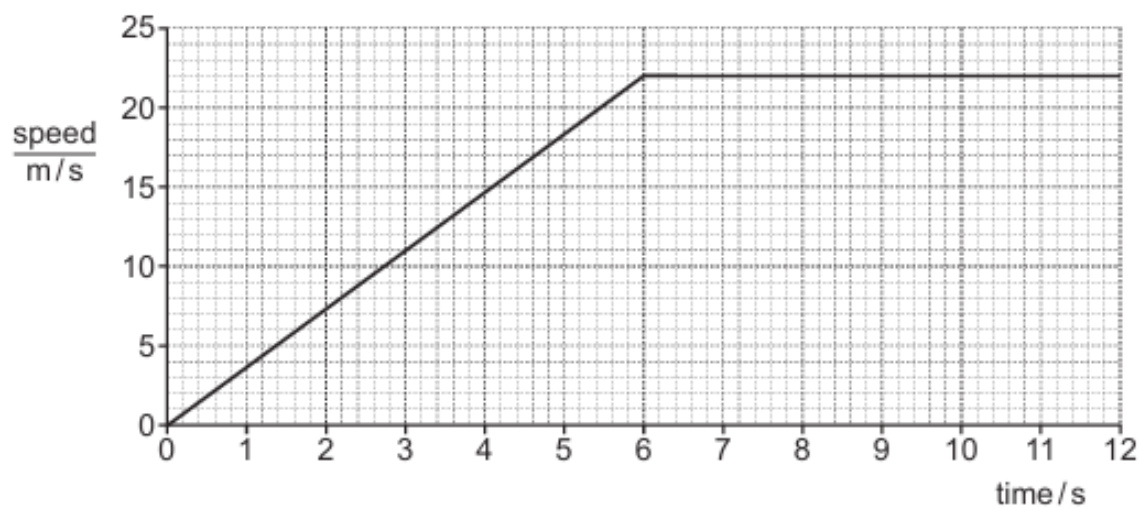


Fig. 1.2

(a) Describe the motion of the glider in the first 12 s.

.....

.....

..... [2]

(b) In the first 6.0 s of the motion, there is a resultant force of 1800 N on the glider.

Using the increase in speed in the first 6.0 s, calculate the mass of the glider.

mass = kg [3]

(c) Determine the distance travelled by the glider in the first 6.0 s of its motion.

distance = m [2]

2. M/J/23/22/Q1

Fig. 1.1 shows the speed–time graph for a car travelling on a straight horizontal road.

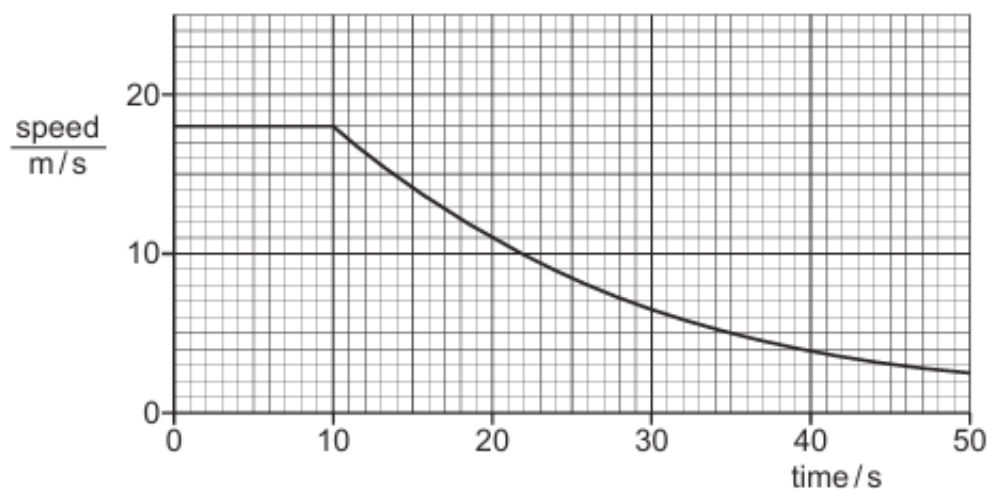


Fig. 1.1

- (a) Describe the motion of the car shown in Fig. 1.1.

.....

.....

.....

..... [2]

- (b) At time $t = 10$ s the engine of the car is switched off. The brakes are not applied.

- (i) Name **two** forces that act on the car to cause the change in motion after $t = 10$ s.

1

2 [1]

- (ii) Suggest why Fig. 1.1 is a curve after $t = 10$ s.

.....

.....

..... [1]

3. O/N/22/21/Q7

Fig. 7.1 shows a child sitting on a sledge on a snow-covered hill of constant slope.

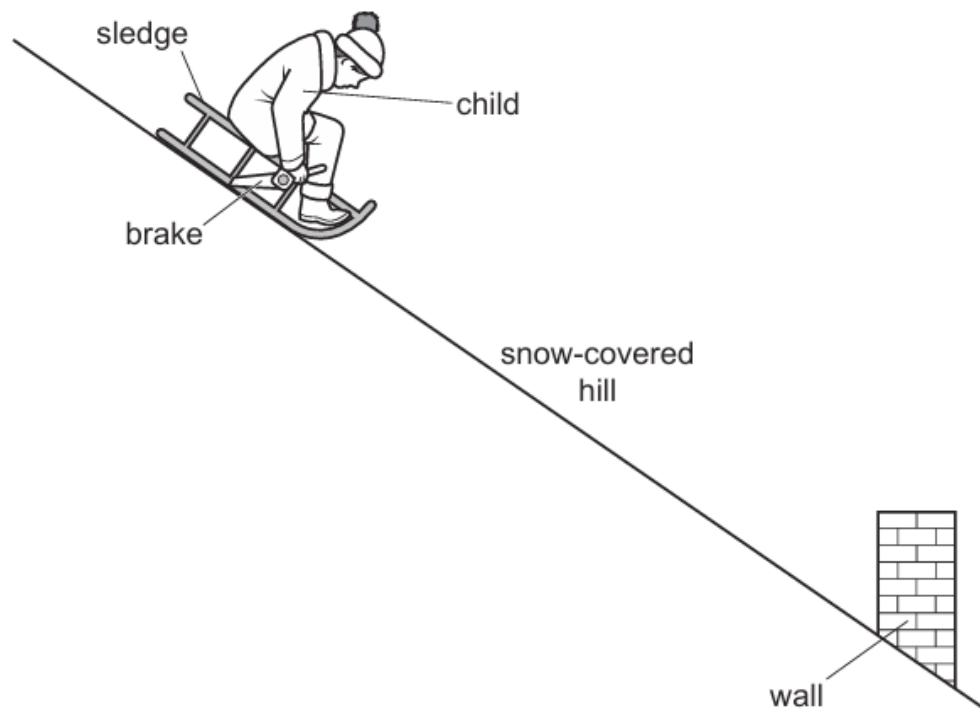


Fig. 7.1 (not to scale)

At time $t = 0$, the child and the sledge begin to move down the hill in a straight line.

When the child sees a wall ahead, he applies the brake.

The child and sledge continue to travel in a straight line until they come to a stop before hitting the wall.

Fig. 7.2 is the speed-time graph for the journey.

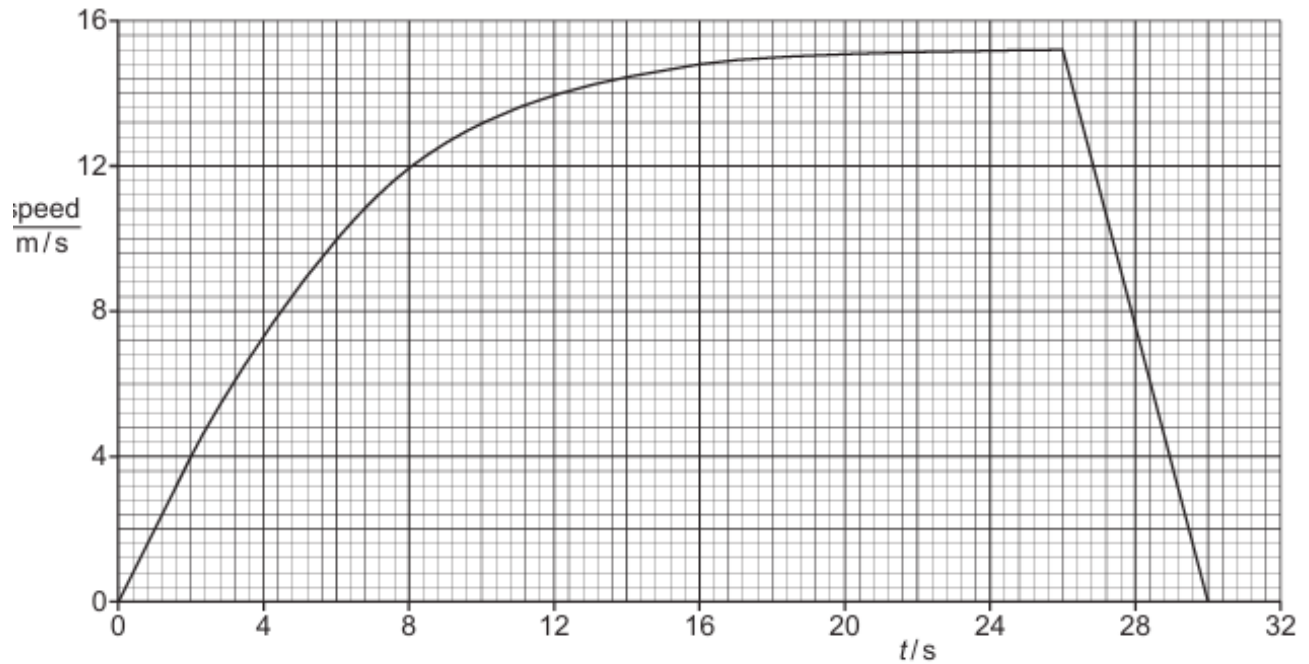


Fig. 7.2

The brake is applied at $t = 26$ s.

(a) Fig 7.2 shows how the speed of the child and sledge varies over the whole of the journey.

Explain why, between $t = 0$ and $t = 26$ s, the speed varies in the way shown by the graph.

.....

.....

.....

.....

.....

..... [3]

(b) At $t = 26$ s, the front of the sledge is 35 m from the wall.

Determine the distance between the front of the sledge and the wall when the sledge stops.

distance = [3]

(c) At $t = 26\text{ s}$, the child and sledge begin to decelerate.

(i) Determine the size of the deceleration.

deceleration = [3]

(ii) The mass of the child is 46 kg and the mass of the sledge is 9.0 kg .

Calculate the resultant force on the child and sledge as they decelerate.

resultant force = [2]

(iii) State the energy transfer that is taking place as the child and sledge decelerate.

.....
.....
..... [2]

(d) At $t = 26\text{ s}$, when the brake is first applied, the child jerks forwards on the sledge.

Explain why.

.....
.....
..... [2]

[Total: 15]

4. O/N/22/22/Q1

A train travels along a straight horizontal track. At time $t = 0$, the train passes through station P at constant speed without stopping.

The driver applies the brakes 70 s before reaching station Q. The train decelerates.

Fig. 1.1 is the speed–time graph for the train from $t = 0$ until it stops at station Q.

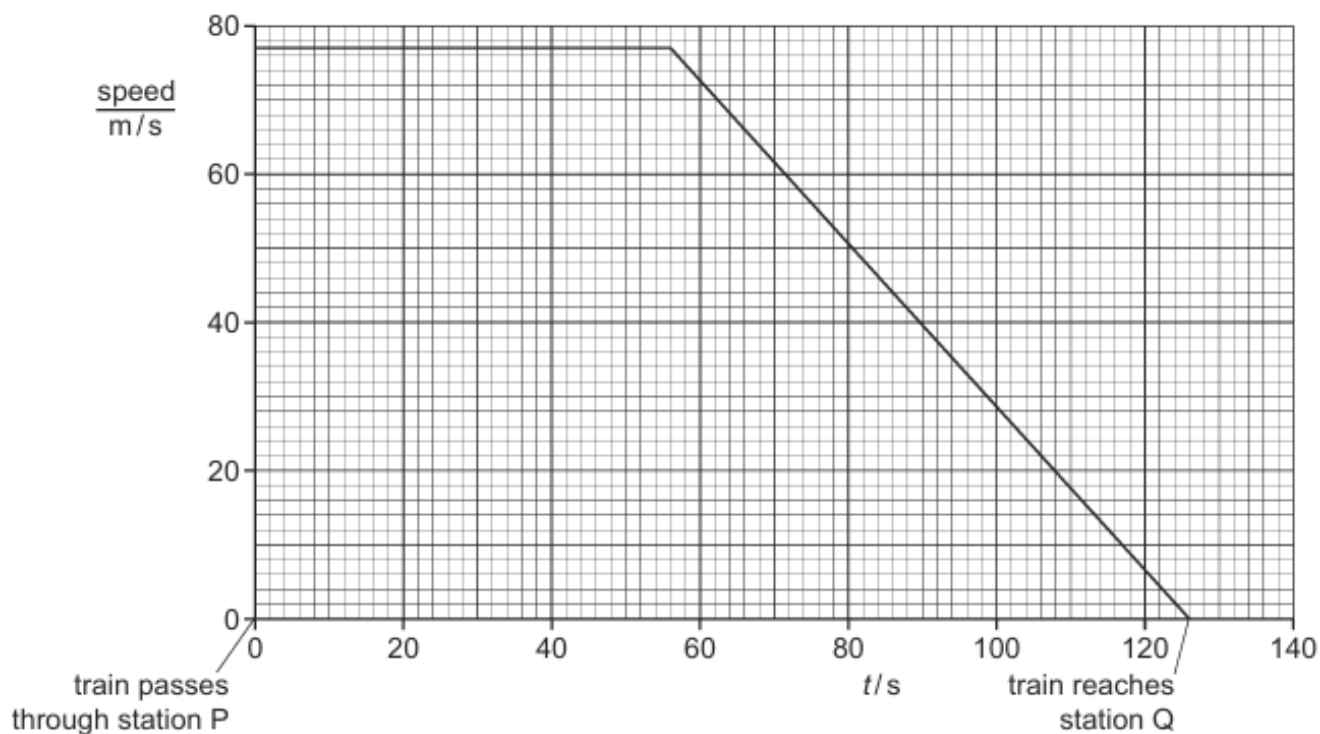


Fig. 1.1

(a) Using Fig. 1.1, determine the distance between station P and station Q.

distance = [3]

(b) The mass of the train is $3.8 \times 10^5 \text{ kg}$.

(i) Determine the deceleration of the train in the 70 s before it stops at station Q.

deceleration = [2]

(ii) Calculate the resultant force on the train as it decelerates.

resultant force = [2]

[Total: 7]

5. M/J/22/22/Q7

Fig. 7.1 shows the speed–time graph for a car travelling on a straight horizontal road.

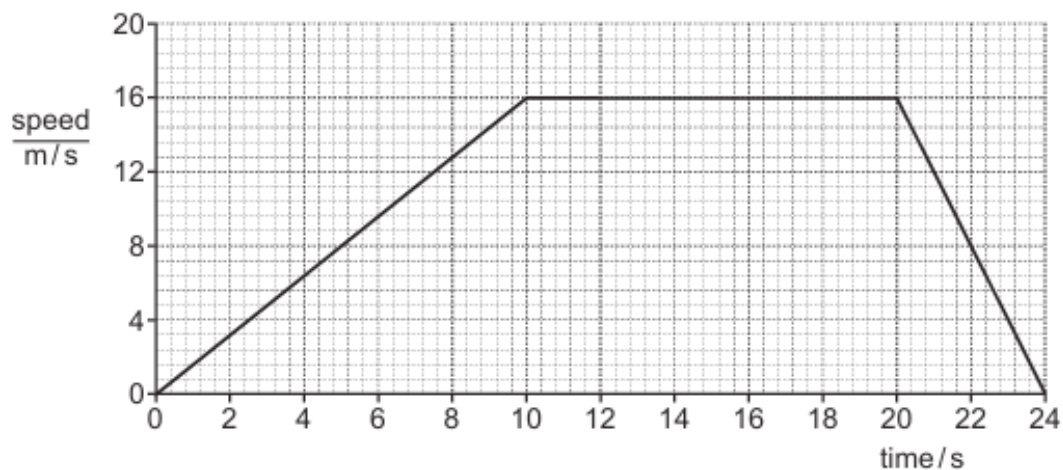


Fig. 7.1

- (a) (i) Describe the motion of the car.

.....

 [3]

- (ii) Using Fig. 7.1, calculate the distance travelled by the car during the 24 s of its motion.

Show your working.

distance = [3]

- (iii) Calculate the average speed of the car during its motion.

average speed = [2]

- (iv) A second car travels at a steady speed. It travels the same distance as the first car in the 24 s of the journey.

On Fig. 7.1, draw the speed–time graph for the second car.

[2]

6. M/J/21/21/Q1

An aircraft flies at a constant height.

Air drag and the force from the aircraft's engines together produce a force on the aircraft of 36 kN due north, as shown in Fig. 1.1.

The wind produces a force of 12 kN towards the east.

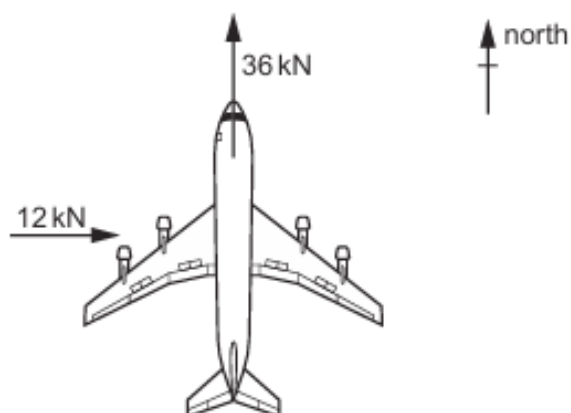


Fig. 1.1 (not to scale)

- (a) Draw a scale drawing to show the resultant force acting on the aircraft.

Use your drawing to determine the size of the resultant force and the angle between the resultant force and north.

size of resultant force =

angle =

[3]

(b) The acceleration of the aircraft is uniform.

(i) Describe how a uniform acceleration differs from a non-uniform acceleration.

.....
.....
.....
..... [2]

(ii) The mass of the aircraft is 60 000 kg.

Calculate the acceleration of the aircraft.

acceleration = [2]

[Total: 7]

7. M/J/21/22/Q2

A car approaches a set of traffic lights. The lights change to red at time $t = 0$.

Fig. 2.1 shows how the speed of the car changes with time.

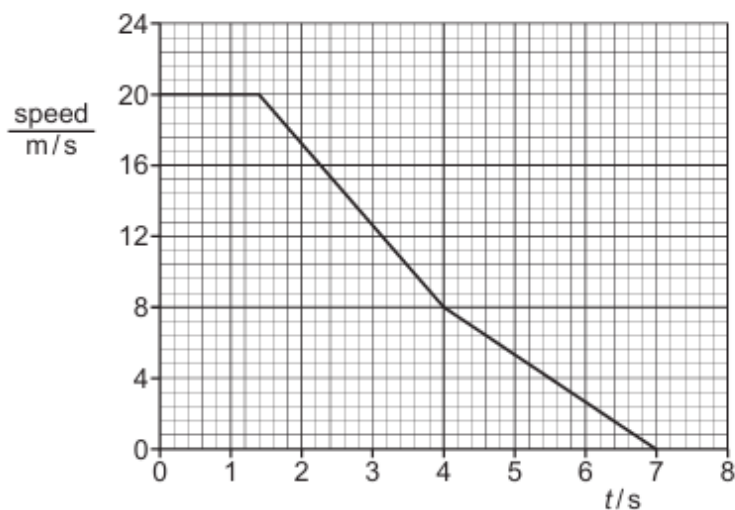


Fig. 2.1

- (a) The car starts to slow down a short time after the lights change to red.

Determine the time between the lights changing to red and the car starting to slow down.

..... [1]

- (b) (i) State what is meant by *uniform acceleration*.

.....
 [1]

- (ii) State how Fig. 2.1 shows that the deceleration of the car between $t = 2$ s and $t = 7$ s is *non-uniform*.

.....
 [1]

- (c) Determine the distance the car travels from the moment the car starts to slow down until it stops.

distance = [3]

[Total: 6]

8. O/N/20/21/Q1

Fig. 1.1 is the distance–time graph for a skydiver who jumps from a balloon at time $t = 0$.

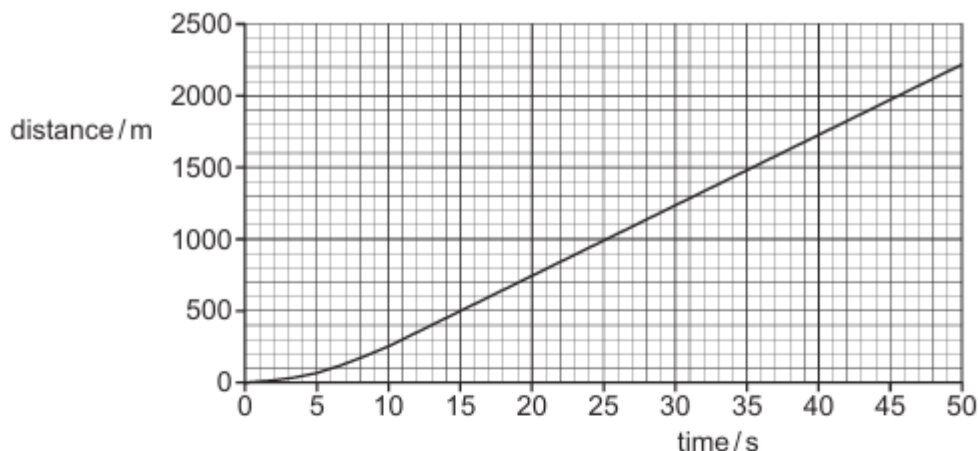


Fig. 1.1

- (a) The first part of the graph shows the motion of the skydiver from when he jumps until he reaches terminal velocity.

- (i) Describe the motion of the skydiver between $t = 0$ and $t = 20$ s.

.....

 [2]

- (ii) Explain the motion of the skydiver between $t = 0$ and $t = 20$ s in terms of the forces acting on him.

.....

 [3]

- (b) Using Fig. 1.1, determine the terminal velocity of the skydiver.

On Fig. 1.1, indicate any values used for your calculation.

terminal velocity = [3]

[Total: 8]

9. O/N/20/22/Q7

A bus leaves a bus-stop at time $t = 0$ and travels along a horizontal road until it reaches a second bus-stop. Fig. 7.1 is the distance-time graph for the bus between $t = 0$ and $t = 60$ s.

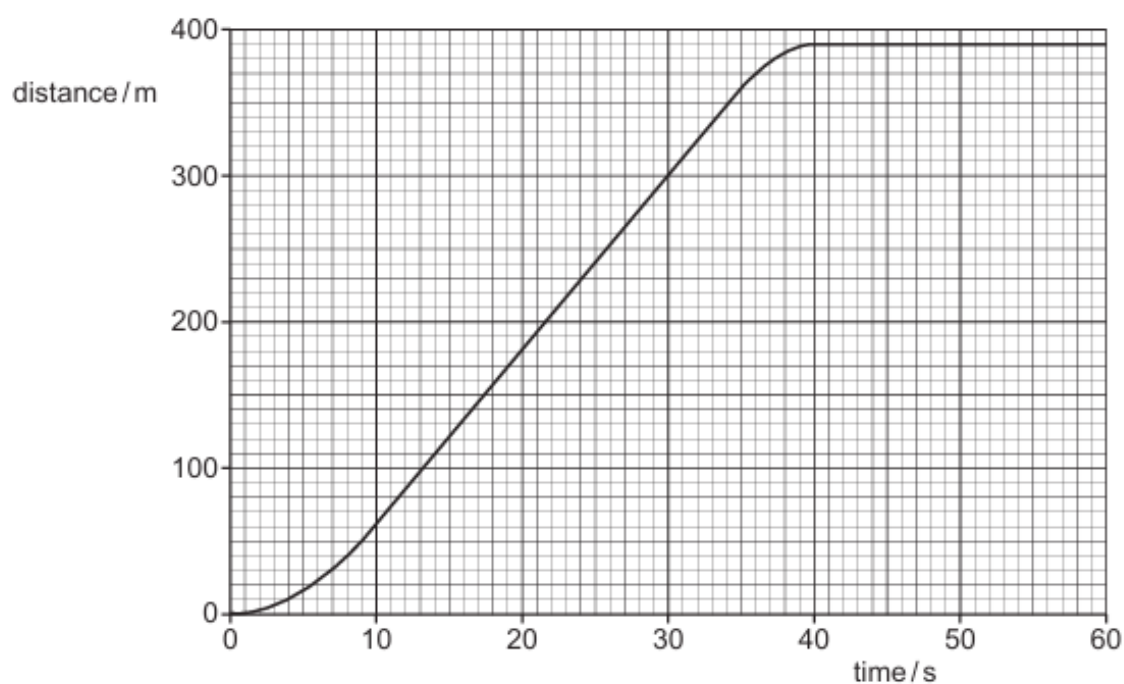


Fig. 7.1

The road on which the bus is travelling is straight except for a short, curved section. The bus travels around this circular curve between $t = 21$ s and $t = 24$ s.

- (a) Describe how the motion of the bus between $t = 0$ and $t = 10$ s differs from its motion between $t = 35$ s and $t = 40$ s.

.....

.....

.....

.....

..... [3]

(b) Determine:

(i) the maximum speed of the bus during these 60 s

maximum speed = [3]

(ii) the average speed of the bus between leaving the first bus-stop and arriving at the second bus-stop.

average speed = [2]

10. M/J/20/22/Q1

Fig. 1.1 shows part of the speed–time graph for an athlete in a race.

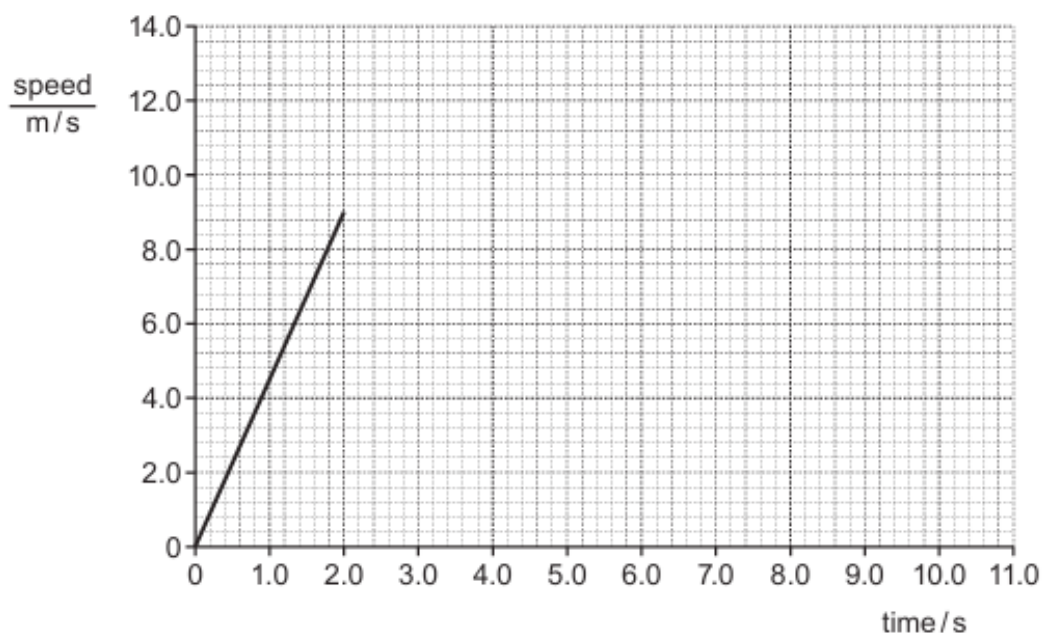


Fig. 1.1

- (a) During the race, the acceleration of the athlete is uniform in the first 2.0 s.

State how the graph shows that the acceleration is uniform.

.....
 [1]

- (b) Determine the distance travelled by the athlete in the first 2.0 s.

distance = [2]

- (c) During the rest of the race:

- from 2.0 s to 5.5 s, the acceleration of the athlete decreases
- at 5.5 s, the athlete reaches a maximum speed of 12 m/s
- from 5.5 s to 8.0 s, the athlete travels at a speed of 12 m/s
- from 8.0 s to 11.0 s, the athlete decelerates, finishing the race at a speed of 10 m/s.

On Fig. 1.1, complete the speed–time graph for times between 2.0 and 11.0 s. [3]

[Total: 6]

11. O/N/19/22/Q1

There is no atmosphere on the Moon.

An astronaut on the Moon drops a feather and a hammer from the same height at the same time. They both accelerate downwards at 1.6 m/s^2 and they hit the ground at the same time.

(a) The weight of the hammer is much larger than that of the feather.

Explain, in terms of their weights and masses, why their accelerations are equal.

.....

.....

.....

.....

.....

..... [3]

(b) Both the feather and the hammer take 1.5 s to fall to the ground from rest.

(i) Calculate the speed of the objects as they hit the ground.

speed = [2]

(ii) On Fig. 1.1, draw the speed-time graph for the fall. At the correct position on the y-axis, write the value of the speed at time $t = 1.5$ s.

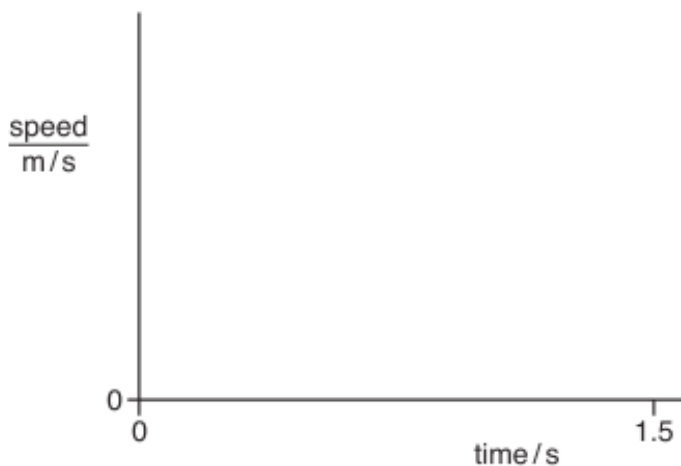


Fig. 1.1

[1]

(iii) Using the speed-time graph in (b)(ii), determine the height from which the objects are dropped.

height = [2]

[Total: 8]

12. M/J/19/21/Q1

Fig. 1.1 shows the distance-time graph for a journey made by a cyclist between town A and town B.

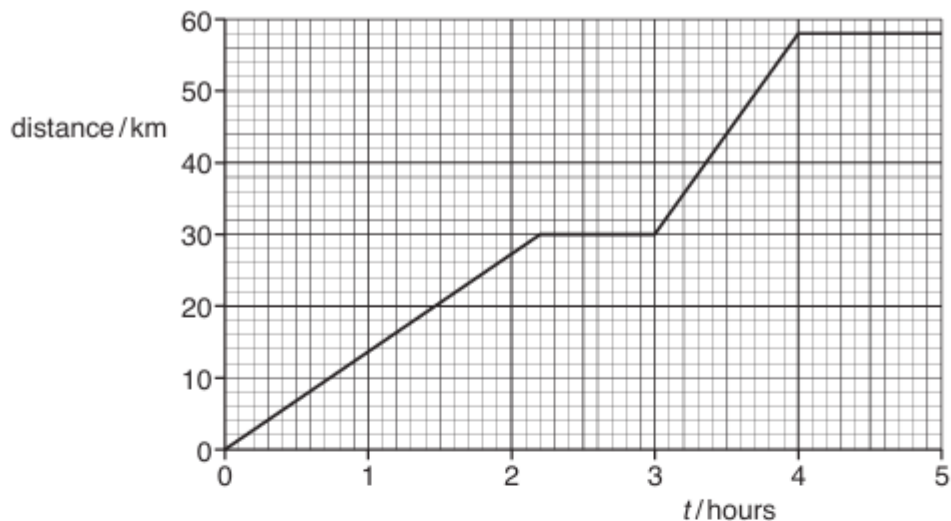


Fig. 1.1

The cyclist leaves town A at time $t = 0$ and arrives at town B at $t = 4.0$ hours.

- (a) (i) Determine the distance between the two towns.

distance = [1]

- (ii) Determine the time for which the cyclist rests during the journey, before he reaches town B.

time = [1]

- (b) The speed of the cyclist near the end of the journey is greater than the speed at the beginning.

State how the graph shows this.

.....
 [1]

- (c) Calculate the average speed of the cyclist for the journey from A to B.

average speed = [2]

[Total: 5]

13. M/J/19/22/Q9

Fig. 9.1 shows a map of the route taken by a car as it travels from town A to town B.

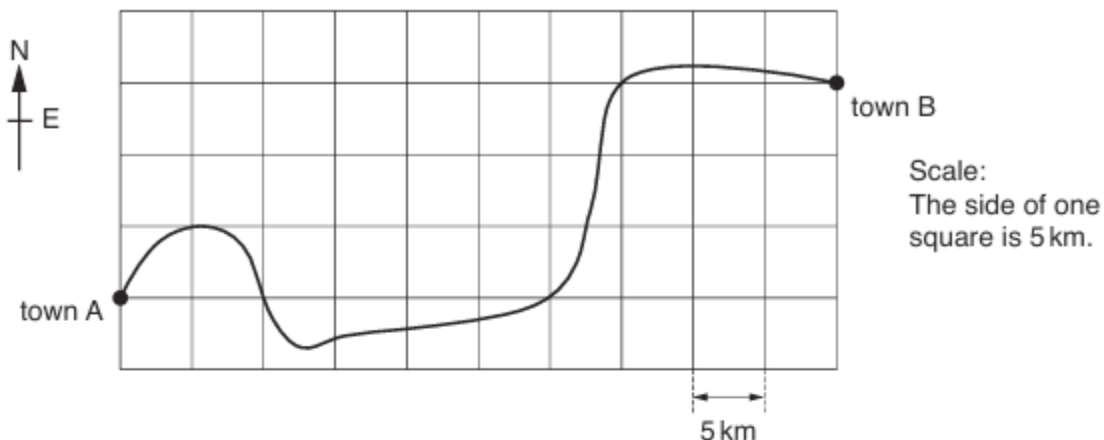


Fig. 9.1

The journey involves a distance and a displacement.

- (a) (i) Using the scale on Fig. 9.1, determine the displacement in travelling from A to B.

size of displacement =

direction of displacement =

[3]

- (ii) State how *distance* differs from *displacement*.

.....

 [1]

- (iii) In some parts of the journey, the car is travelling at a constant speed but is also accelerating.

Explain how this is possible.

.....

 [2]

14. M/J/18/22/Q1

A student drops a small metal object into a cylinder of oil. The object falls alongside a vertical ruler and a camera records its position at 1.0 s intervals, as shown in Fig. 1.1.

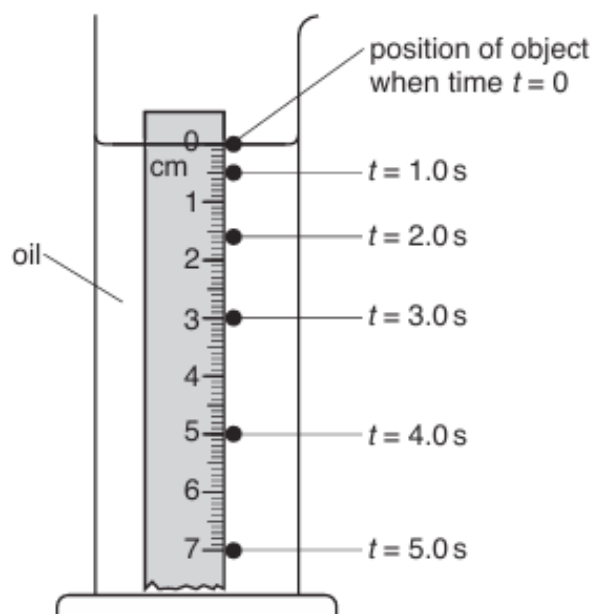


Fig. 1.1

(a) On the grid in Fig. 1.2, plot a distance-time graph for the object.

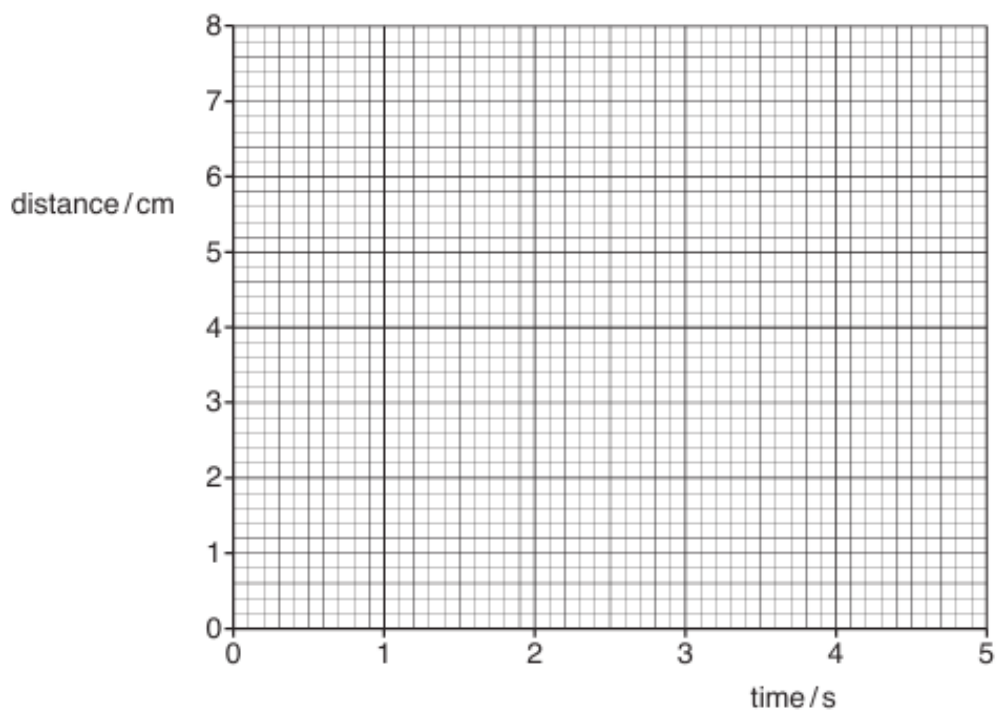


Fig. 1.2

[2]

(b) The object has a constant speed for some of the time.

(i) State and explain how a distance-time graph shows that speed is constant.

.....

.....

.....

.....[2]

(ii) Explain, in terms of the forces acting on the object, how it is able to fall at a constant speed.

.....

.....

.....

.....[2]

15. M/J/18/21/Q1

A cyclist travels along a straight, horizontal road, as shown in Fig. 1.1.



Fig. 1.1

- (a) Describe how the average speed of the cyclist can be measured.

In your account, suggest what equipment is used, the measurements that are made and how the average speed is calculated.

.....

.....

.....

.....

.....

.....[3]

- (b) For part of the journey, the cyclist has a constant, negative acceleration.

- (i) State what is meant by a *negative acceleration*.

.....

.....[1]

- (ii) The size of the acceleration is 1.8 m/s^2 .

Calculate the change in speed of the cyclist in a time of 2.5 s.

change in speed =[2]

16. O/N/18/21/Q1

A girl of mass 35 kg, on a bicycle, accelerates from rest and travels down a slope in a straight line. The girl does not use the pedals.

Fig. 1.1 shows that the gradient of the slope is constant.

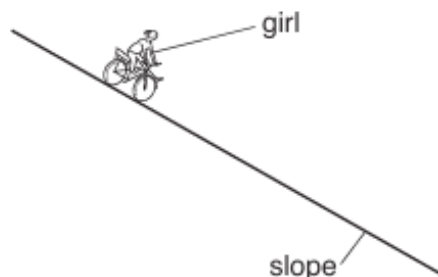


Fig. 1.1

- (a) Calculate the resultant force on the girl when she is accelerating at 2.6 m/s^2 .

resultant force = [2]

- (b) At first, her acceleration is constant. At time t_1 , her acceleration starts to decrease gradually until she is travelling at a constant speed in a straight line.

- (i) On Fig. 1.2, sketch a speed-time graph for the girl from when she starts moving until she is travelling at a constant speed. [2]

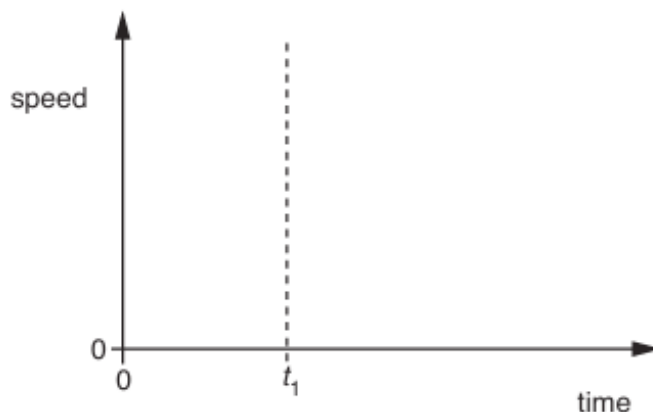


Fig. 1.2

- (ii) State how the distance travelled is found from a speed-time graph.

.....
 [1]

17. M/J/17/22/Q1

A car accelerates from rest in a straight line. During the first 14 s, the acceleration is uniform and the car reaches a speed of 25 m/s.

- (a) (i) Calculate the acceleration of the car.

acceleration =[2]

- (ii) After the first 14 s, the speed of the car continues to increase but the acceleration decreases. From 70 s to 80 s after the start, the car moves at a constant speed of 55 m/s.

On Fig. 1.1, draw a possible speed-time graph for the car.

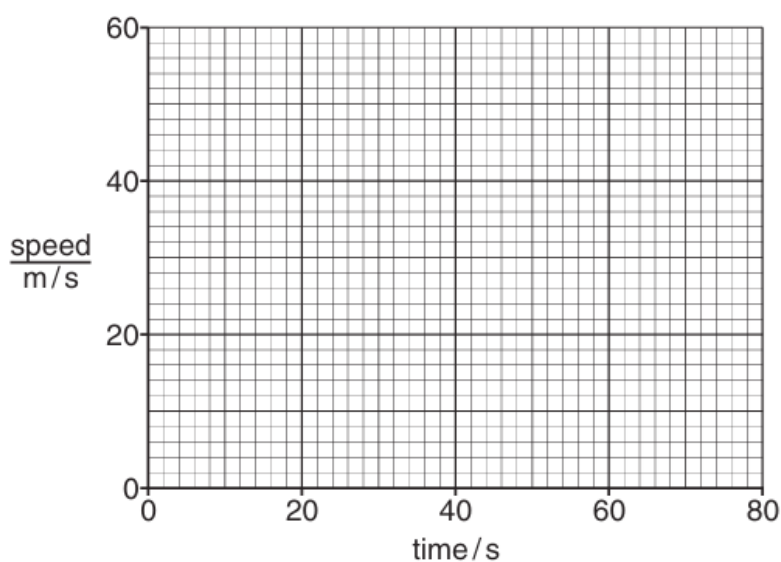


Fig. 1.1

[2]

18. O/N/16/22/Q1

A skier sets off from rest and accelerates uniformly at 3.4 m/s^2 in a straight line for 5.0 s.

- (a) Calculate the speed of the skier after 5.0 s.

speed =[2]

- (b) At 5.0 s, the skier stops accelerating and travels on for a further 10.0 s at a constant speed.

- (i) State the size of the resultant force acting on the skier during these 10.0 s.

.....[1]

- (ii) On Fig. 1.1, sketch a speed-time graph for the skier during the whole 15.0 s. [3]

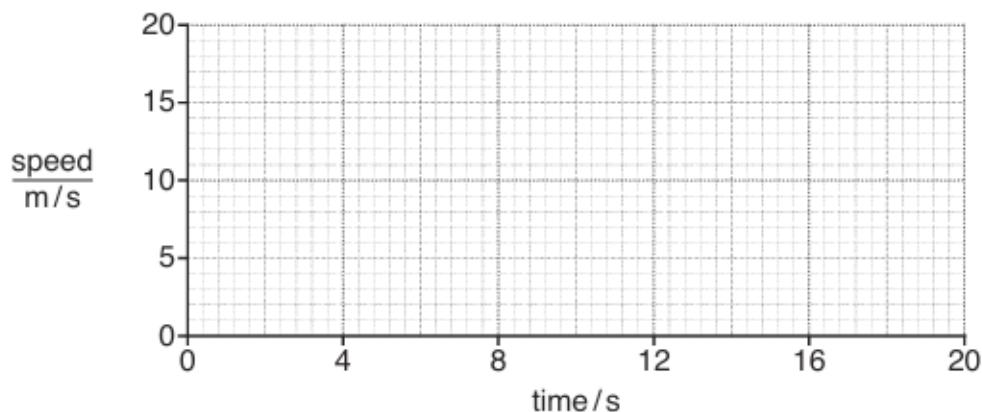


Fig. 1.1

- (iii) State how the distance travelled by the skier can be determined using the speed-time graph.

.....
.....[1]

19. O/N/15/21/Q1

Fig. 1.1 shows an archer firing an arrow at a target during a competition.

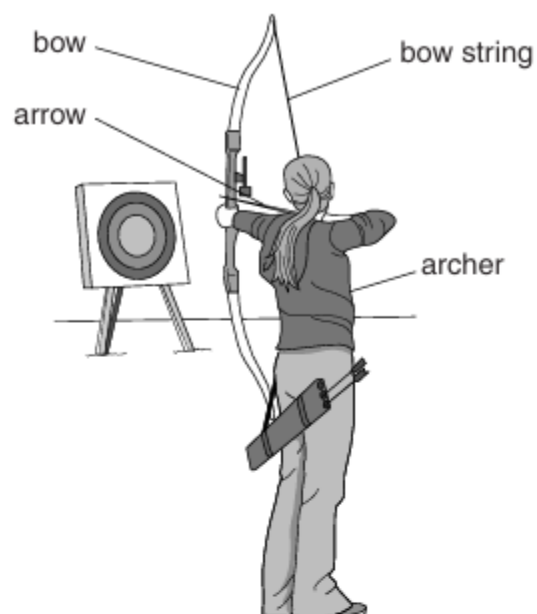


Fig. 1.1

The arrow has a mass of 0.018 kg and is initially at rest. The arrow leaves the bow 0.011 s after the bow string is released. When it leaves the bow, the velocity of the arrow is 95 m/s .

(a) Calculate

(i) the average acceleration of the arrow during the 0.011 s ,

average acceleration = [2]

(ii) the average force exerted on the arrow during the 0.011 s .

average force = [2]

(b) In practice, the force exerted on the arrow gradually decreases to zero during firing.

On Fig. 1.2, sketch a possible speed-time graph for the arrow as it is being fired.

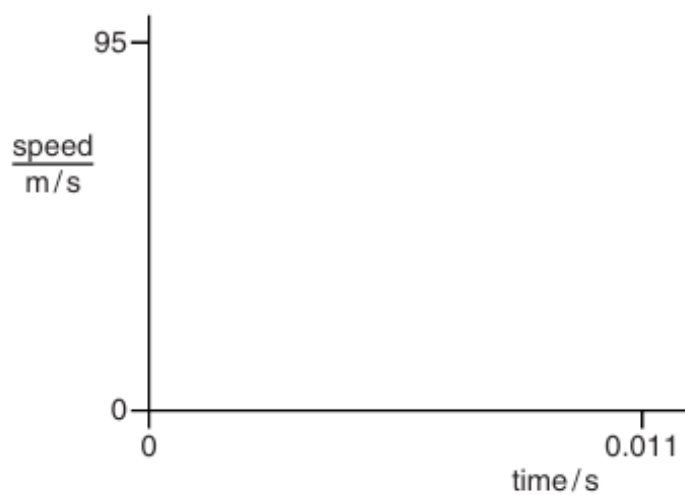


Fig. 1.2

[2]

20. M/J/15/21/Q1

Fig. 1.1 shows the distance-time graph for two cyclists A and B. They start a 500 m race together but finish the race at different times.

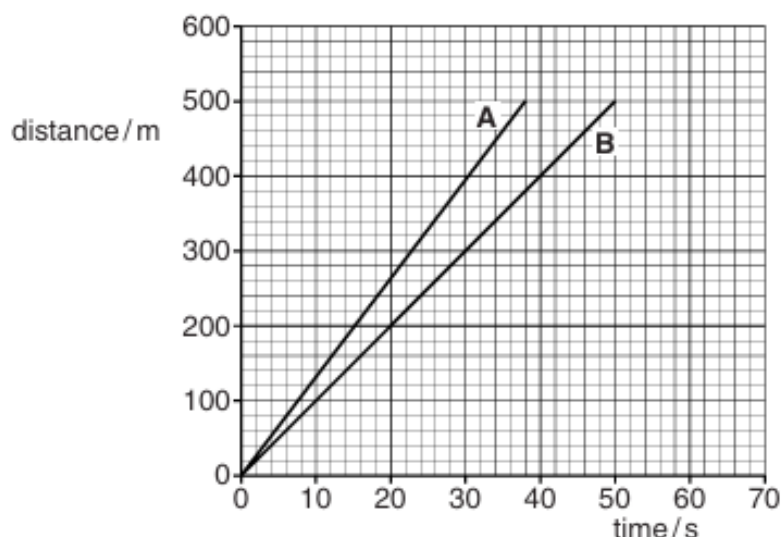


Fig. 1.1

(a) Use Fig. 1.1 to determine

(i) the distance between A and B at time $t = 20$ s,

.....[1]

(ii) the difference in the time taken by A and B for the race.

.....[1]

(b) Cyclist C starts the race at the same time as A and B and covers the first 200 m of the race at a constant speed of 5.0 m/s. He then accelerates and finishes the race at $t = 60$ s.

(i) On Fig. 1.1, draw the distance-time graph for cyclist C.

[2]

(ii) Calculate the average speed of cyclist C for the whole race.

speed =[2]

21. M/J/15/22/Q11

A bungee jumper falls from a bridge above a river, as shown in Fig. 11.1.

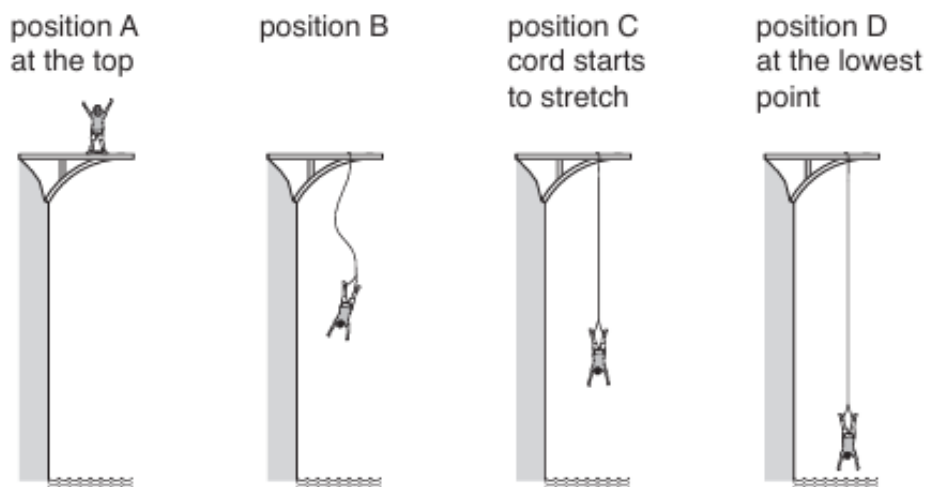


Fig. 11.1 (not to scale)

The man starts from position A in Fig. 11.1. The elastic cord starts to stretch at position C and he stops for the first time at position D. He continues to rise and fall.

Fig. 11.2 shows how the velocity of the man varies with time t .

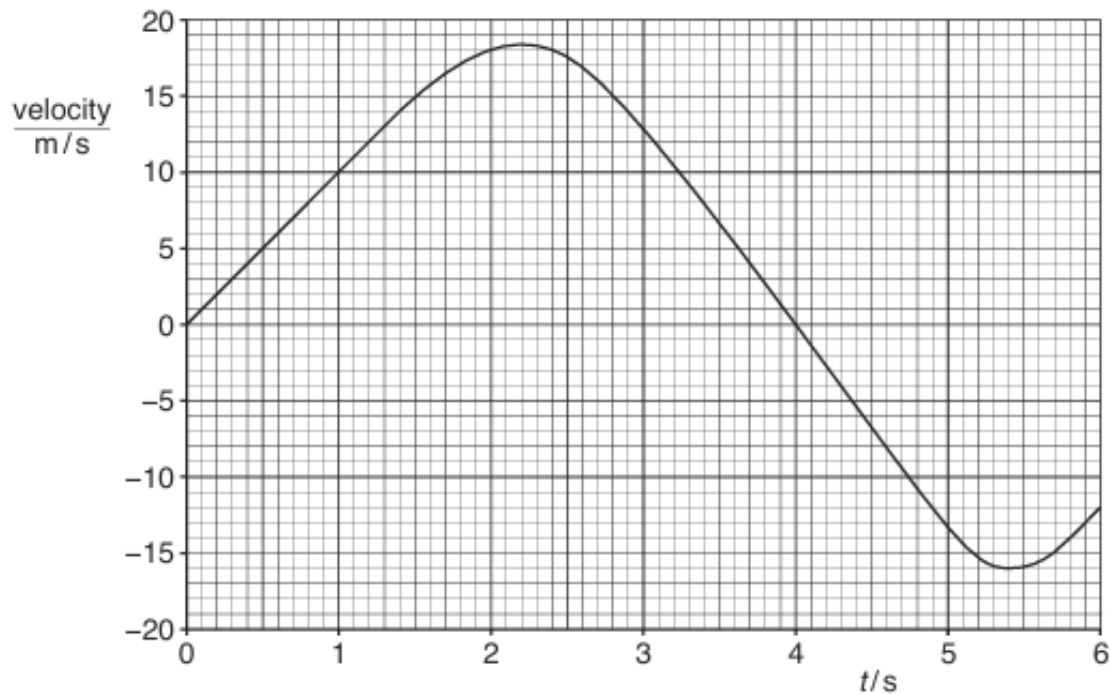


Fig. 11.2

(a) (i) State what is meant by *velocity*.

.....
..... [2]

(ii) State the difference between a positive velocity and a negative velocity.

.....
..... [1]

(iii) In the first 1.4 s the acceleration is uniform.

1. Using values from Fig. 11.2, determine the acceleration of the man in the first 1.4 s.

acceleration = [3]

2. Comment on your value of acceleration.

.....
..... [1]

22. O/N/14/22/Q9

(a) Explain what is meant by *uniform acceleration*.

.....

[2]

(b) Acceleration is a vector quantity.

State how a vector quantity differs from a scalar quantity.

.....
[1]

(c) A rock from space is travelling in a straight line at high speed when it enters the Earth's atmosphere. Fig. 9.1 is the speed-time graph for the rock from time $t = 0$ to time $t = 50$ s.

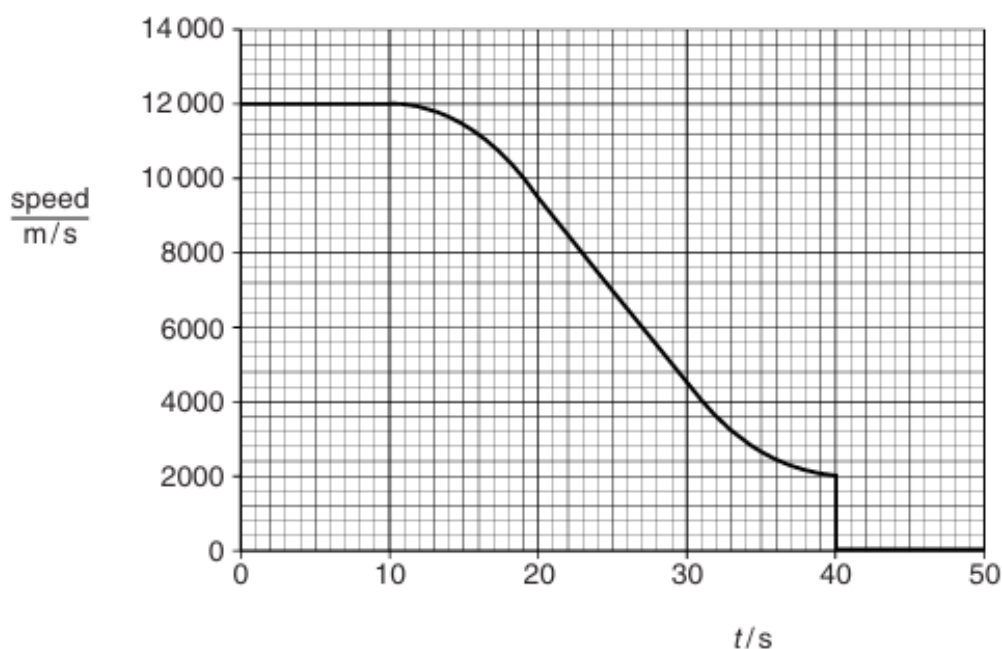


Fig. 9.1

(i) On Fig. 9.1, mark

- a letter X, where the rock is moving with a constant speed,
- a letter Y, where the rock has a uniform deceleration,
- a letter Z, where the rock has a non-uniform deceleration.

[3]

- (ii) At time $t = 25$ s, the mass of the rock is 8.4 kg. For the time $t = 25$ s, determine the size of the acceleration of the rock,

acceleration =[3]

23. M/J/14/22/Q9

A children's ride consists of a steel cable that runs between two posts of different heights, as shown in Fig. 9.1.

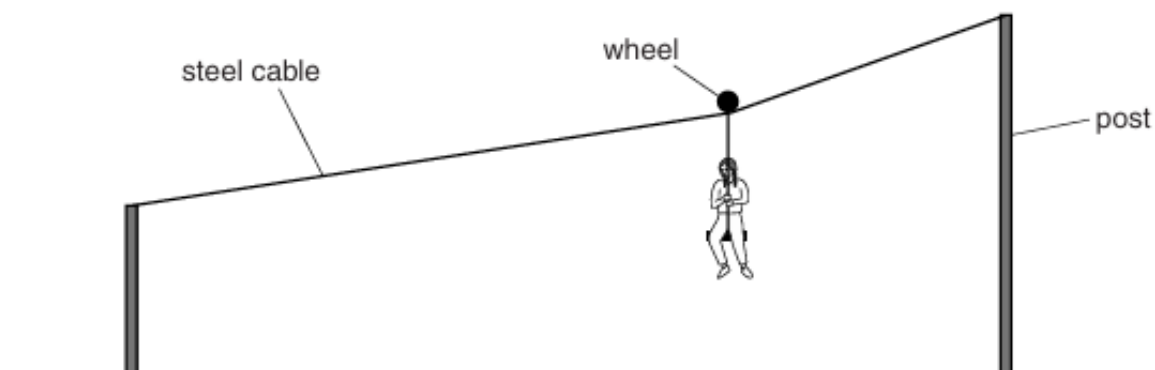


Fig. 9.1

A girl starts and finishes the ride at rest. Her horizontal motion can be taken as

- an initial uniform acceleration for 3.0 s, followed by
- a constant speed of 2.4 m/s for a further 5.0 s and
- a final uniform deceleration that lasts for 1.0 s.

(a) On Fig. 9.2, draw a speed-time graph of the horizontal motion.

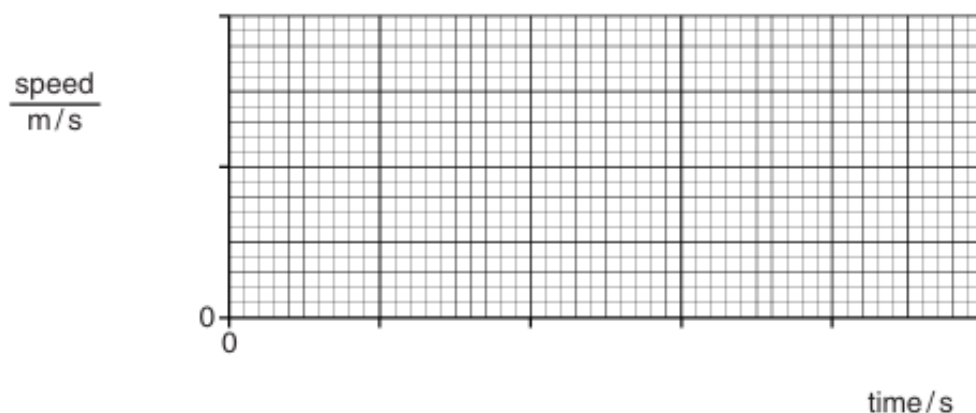


Fig. 9.2

[3]

(b) Explain what is meant by *uniform acceleration*.

.....

 [2]

- (c) The final deceleration is larger in size than the initial acceleration.

Explain how the data shows this.

.....
..... [1]

- (d) Calculate the horizontal distance travelled by the girl in the first 8.0 s.

distance = [3]

- (e) (i) The girl has a mass of 30 kg and falls a vertical distance of 1.6 m during the ride.

The gravitational field strength g is 10 N/kg.

Calculate the decrease in gravitational potential energy of the girl.

decrease in potential energy = [2]

- (ii) The gain in kinetic energy of the girl is less than the decrease in her potential energy. Suggest one reason for this.

.....
..... [1]

- (f) A group of pupils make measurements to show that the girl's speed is constant during the middle section of the ride.

Suggest what measurements are made and how they show that the speed is constant.

.....
.....
.....
.....
.....
..... [3]

24. O/N/13/22/Q10

Fig. 10.1 shows an electric train of mass $5.5 \times 10^5 \text{ kg}$.



Fig. 10.1

The train is initially at rest. The electric motor exerts a constant force and the train accelerates. Its acceleration decreases until the train reaches a speed of 40 m/s .

The train then continues at this constant speed.

- (a) (i)** On the axes in Fig. 10.2, sketch a speed-time graph for the train.

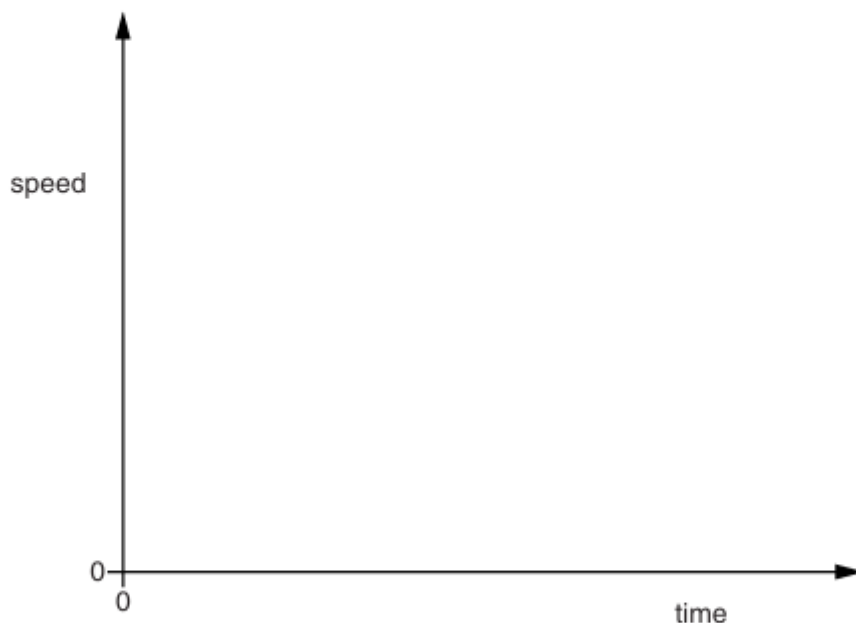


Fig. 10.2

[3]

- (ii)** Describe how the speed-time graph for the train is used to determine the distance travelled by the train when it is accelerating.

.....
.....
.....[2]

25. M/J/13/22/Q1

A ball rolls down a slope, as shown in Fig. 1.1.

The metre rule shows the position of the ball at times $t = 0$, 1.0 s, 2.0 s and 3.0 s.

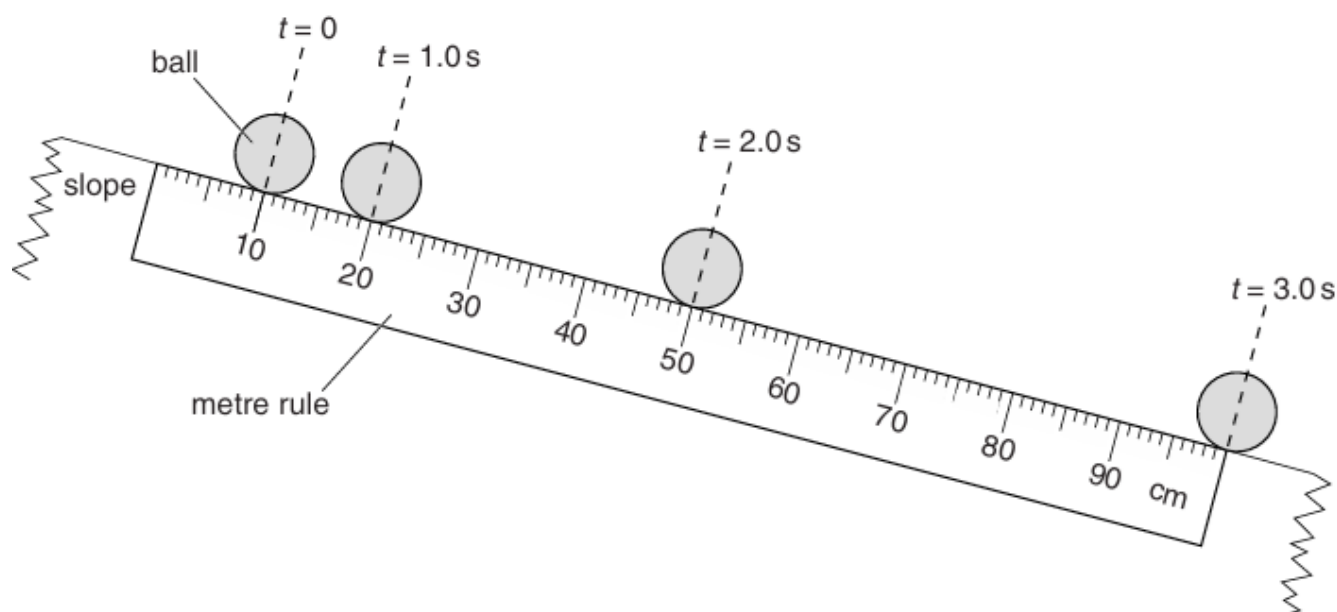


Fig. 1.1

(a) Explain how Fig. 1.1 shows that the ball is accelerating.

.....
..... [1]

(b) Calculate the average speed of the ball between $t = 1.0$ s and 3.0 s.

average speed = [2]

26. O/N/11/22/Q9

Fig. 9.1 is the speed-time graph for a racing car of total mass 650 kg as it sets off from rest at the start of a race.

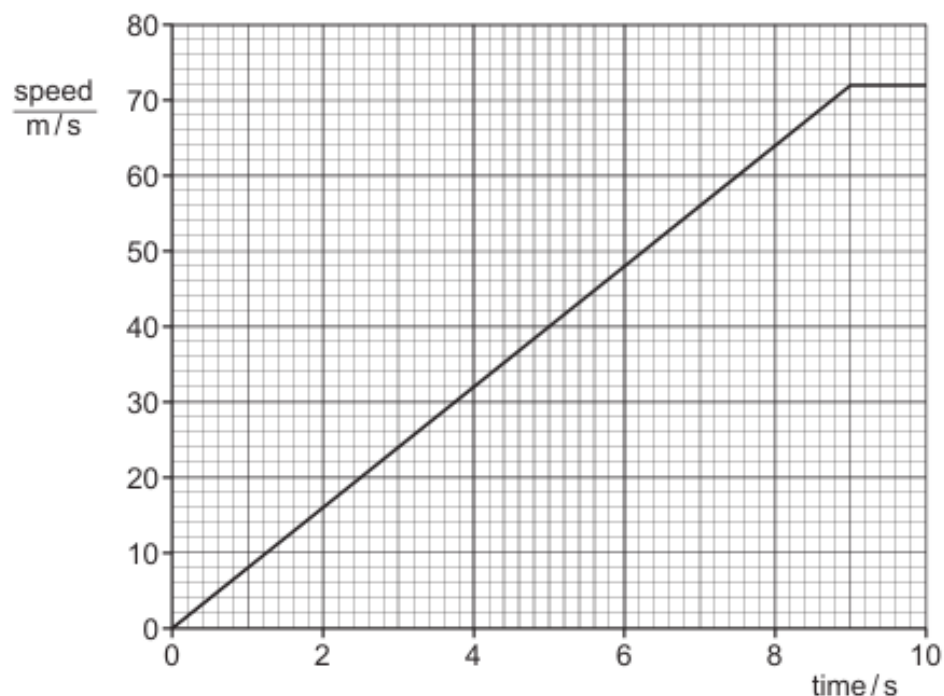


Fig. 9.1

The car travels in a straight line until time $t = 9.0$ s.

(a) State the speed of the car at time $t = 9.0$ s.

speed =[1]

(b) Calculate, for the car between times $t = 0$ and $t = 9.0$ s,

(i) the distance travelled,

distance =[2]

(ii) the acceleration of the car,

acceleration =[2]

27. M/J/11/22/Q9

Fig. 9.1 shows a sky-diver falling vertically.



Fig. 9.1

The sky-diver starts from rest at time $t = 0$.

His acceleration is non-uniform until he reaches a steady speed of 50 m/s at $t = 10 \text{ s}$.

He opens his parachute at $t = 20 \text{ s}$ and decelerates until $t = 25 \text{ s}$.

From $t = 25 \text{ s}$ he falls at a steady speed of 5 m/s .

(a) (i) On Fig. 9.2, draw the speed-time graph for the sky-diver.

[4]

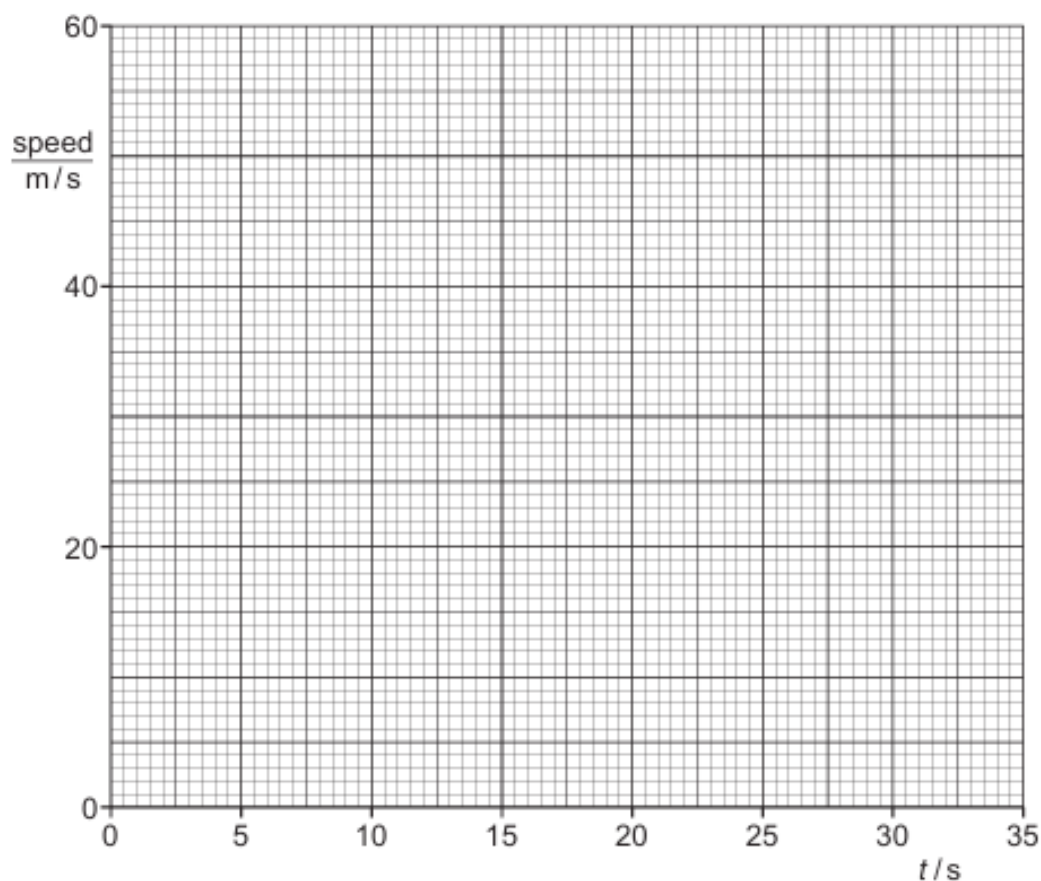


Fig. 9.2

(ii) State how your graph shows that the acceleration is non-uniform between $t = 0$ and $t = 10 \text{ s}$.

.....
[1]

- (b) Explain, in detail, why **after** the sky-diver opens his parachute, he decelerates and eventually reaches a steady speed.

.....

.....

.....

.....

.....

.....[4]

- (c) For the time interval between $t = 10\text{ s}$ and $t = 20\text{ s}$, calculate the distance that the sky-diver falls.

distance =[1]

- (d) The mass of the sky-diver is 60 kg .

For the time interval between $t = 20\text{ s}$ and $t = 25\text{ s}$,

- (i) calculate the average deceleration of the sky-diver,

average deceleration =[2]

- (ii) calculate the average resultant force acting on the sky-diver,

force =[2]

- (iii) state how your graph in Fig. 9.2 may be used to obtain the distance that the sky-diver falls.

.....

.....[1]

28. M/J/11/21/Q1

Fig. 1.1 shows the speed-time graph for a car.

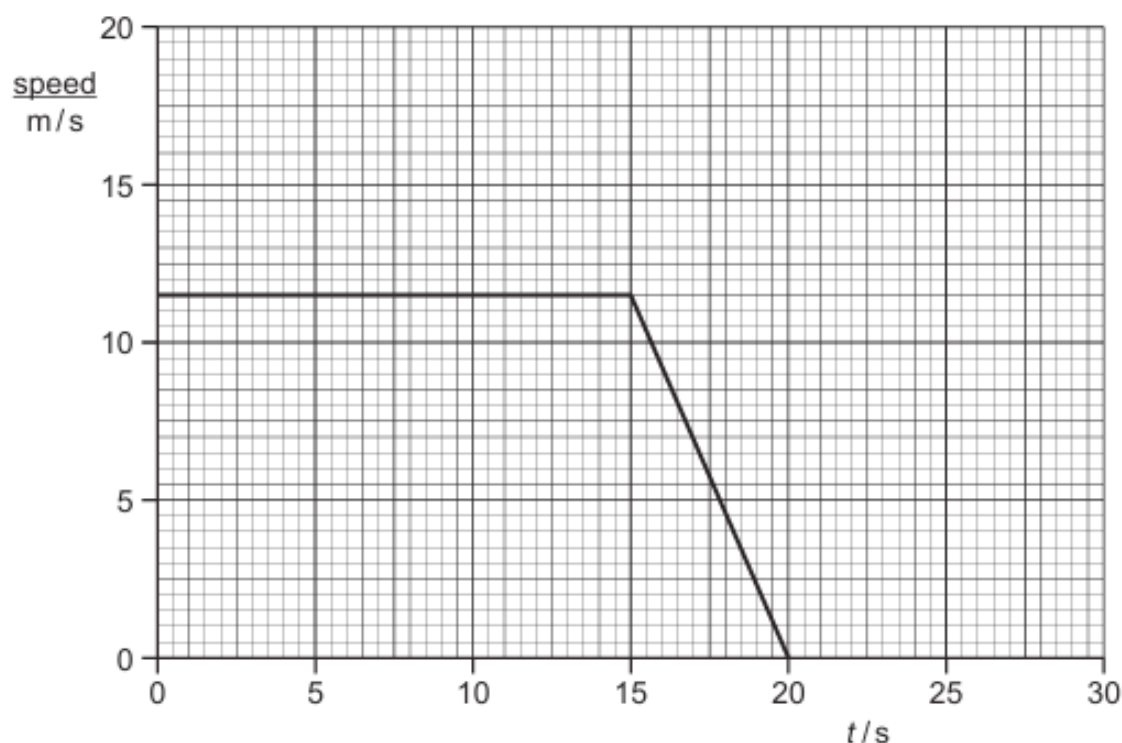


Fig. 1.1

At time $t = 15$ s, the brakes are applied and the car comes to rest with uniform deceleration.

- (a) (i) State the speed of the car just before the brakes are applied.

speed = [1]

- (ii) Explain what is meant by *uniform deceleration*.

.....

 [2]

- (b) A lorry travels at a constant speed of 18 m/s for 15 s. At time $t = 15$ s, the brakes are applied and the lorry slows down with the same deceleration as the car.

- (i) On Fig. 1.1, draw the speed-time graph for the lorry. [2]

- (ii) Explain how your graph shows that, while braking, the lorry travels further than the car.

.....
 [1]

29. O/N/10/22/Q1

Fig. 1.1 shows a cricket ball as it comes into contact with a cricket bat.



Fig. 1.1

The cricket ball has a mass of 0.16 kg and it hits the bat with a speed of 25 m/s. After being in contact with the bat for 0.0013 s, the ball rebounds with a speed of 22 m/s in the direction exactly opposite to its original direction.

(a) State the difference between *speed* and *velocity*.

.....
.....[1]

(b) Calculate

(i) the change in velocity of the cricket ball,

velocity change =[1]

(ii) the average acceleration of the ball whilst it is in contact with the bat,

acceleration =[2]

30. O/N/10/21/Q1

A window cleaner drops a sponge from the balcony of a hotel at time $t = 0$. Fig. 1.1 is the speed-time graph for the motion of the sponge.

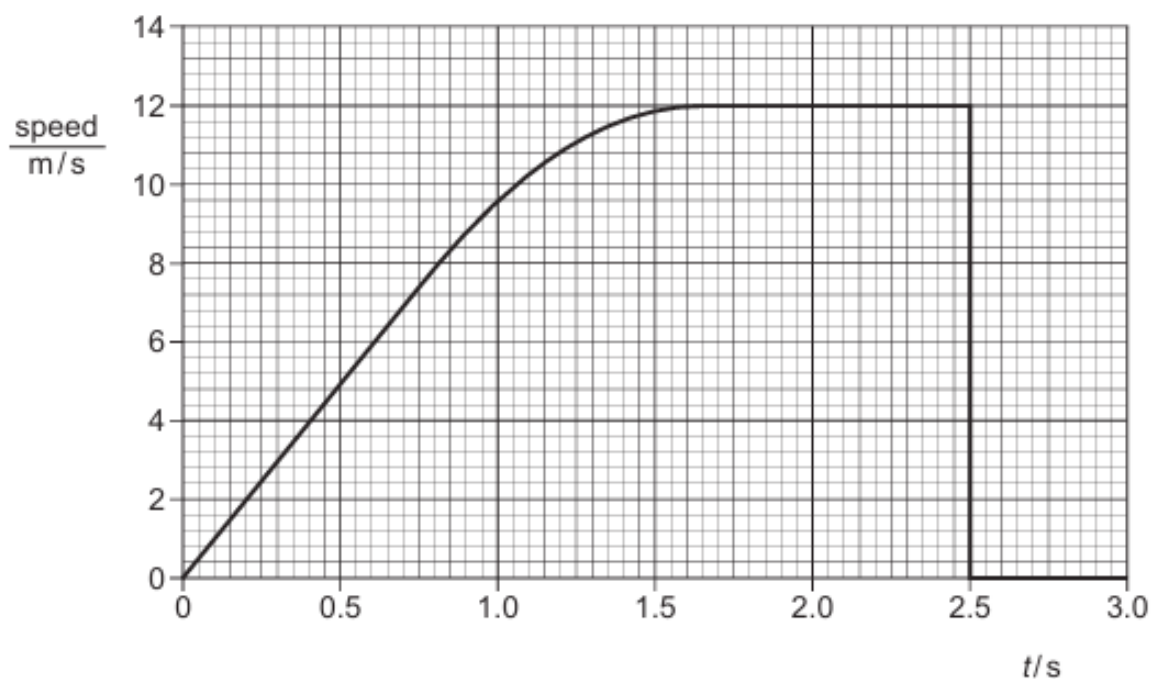


Fig. 1.1

- (a) State a value for t when the sponge is
- (i) moving with a uniform speed,
..... [1]
 - (ii) accelerating at a non-uniform rate,
..... [1]
 - (iii) decelerating.
..... [1]
- (b) Calculate the distance travelled by the sponge between $t = 0$ and $t = 0.75$ s.

distance = [2]

31. M/J/09/Q1

A piece of paper falls from 4.0 m above the ground.

Fig. 1.1 shows how the height h above the ground varies with the time t .

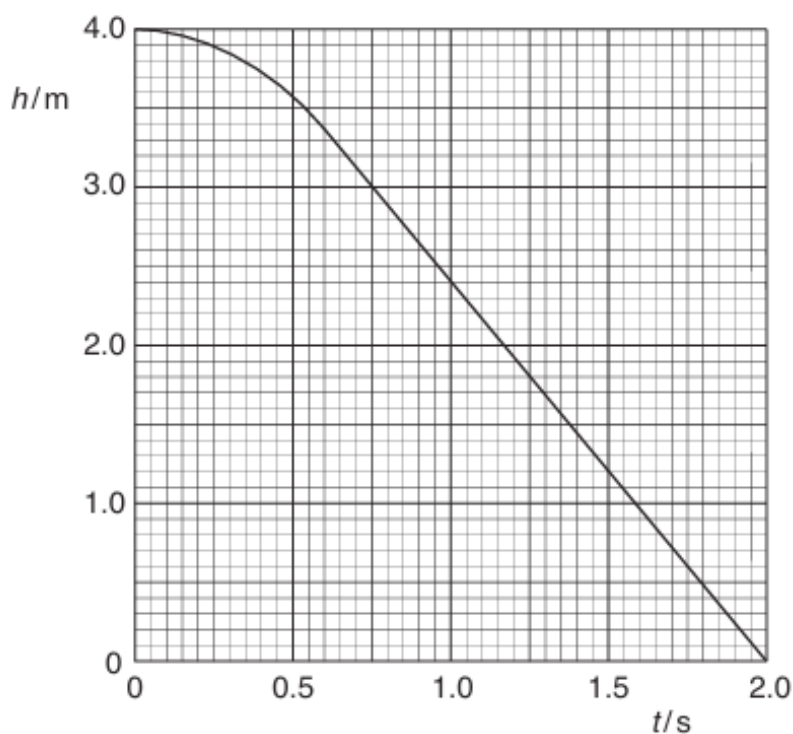


Fig. 1.1

- (a)** State what happens to the speed of the paper as it falls.

.....
.....
..... [2]

- (b)** Calculate the speed of the paper at time $t = 1.5$ s.

speed = [2]

32. O/N07/Q1

Fig. 1.1 is a graph of the vertical speed of the parachutist plotted against time t .

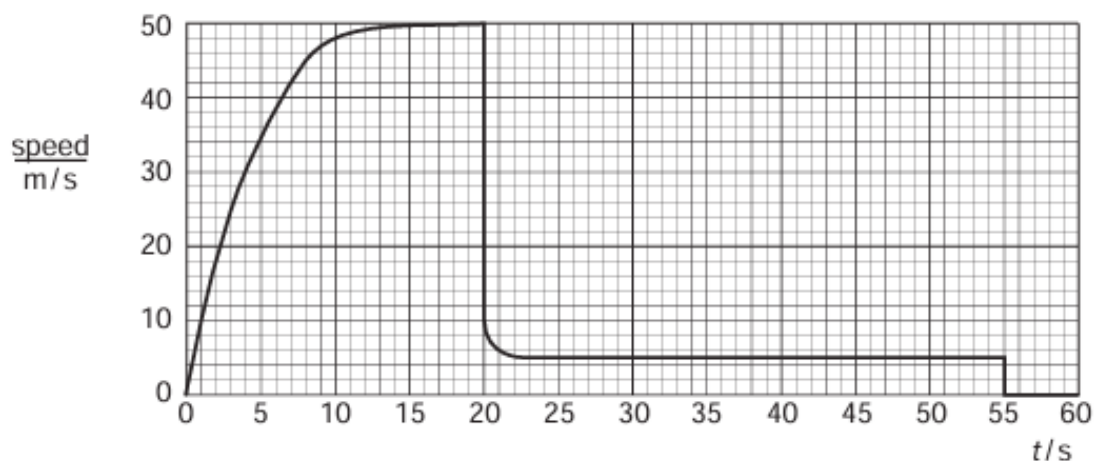


Fig. 1.1

- (a)** State what happens at $t = 20\text{ s}$ and $t = 55\text{ s}$.

at 20 s

at 55 s [1]

- (b)** Describe the motion of the parachutist between $t = 0$ and $t = 20\text{ s}$.

.....

.....

..... [2]

- (c)** Explain, in terms of the forces acting, why the speed of the parachutist is constant between $t = 25\text{ s}$ and $t = 55\text{ s}$.

.....

.....

..... [2]

- (d)** Calculate the distance travelled by the parachutist between $t = 25\text{ s}$ and $t = 55\text{ s}$.

distance = [2]

33. M/J/07/Q1

Two athletes, A and B, run a 100 m race. At time $t = 0$, a gun is fired to start the race.

Fig. 1.1 shows the distance-time graph for the two athletes.

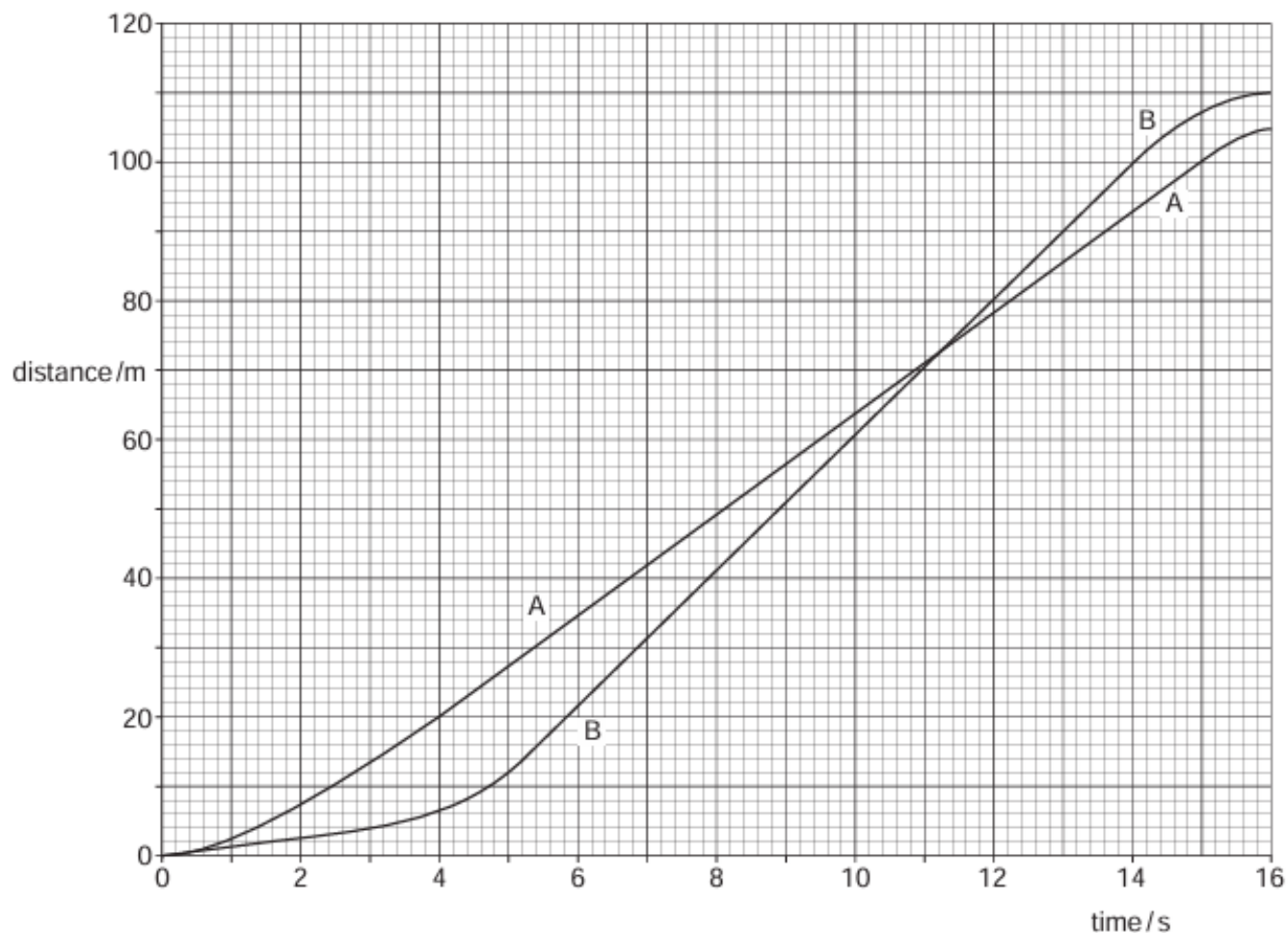


Fig. 1.1

(a) Describe the motion of athlete A during the first 8 s of the race.

.....

.....

.....

.....

..... [2]

(b) State the distance between the two athletes as the winner passes the 100 m mark.

..... [1]

(c) Calculate the speed of athlete A between $t = 4$ s and $t = 15$ s.

speed = [2]

34. O/N/06/Q1

Fig. 1.1 represents the motion of a car along a straight road. As the car approaches a small town, it slows down. The car travels at a constant speed from the start of the town to the end of the town. After passing through the town, the car speeds up.

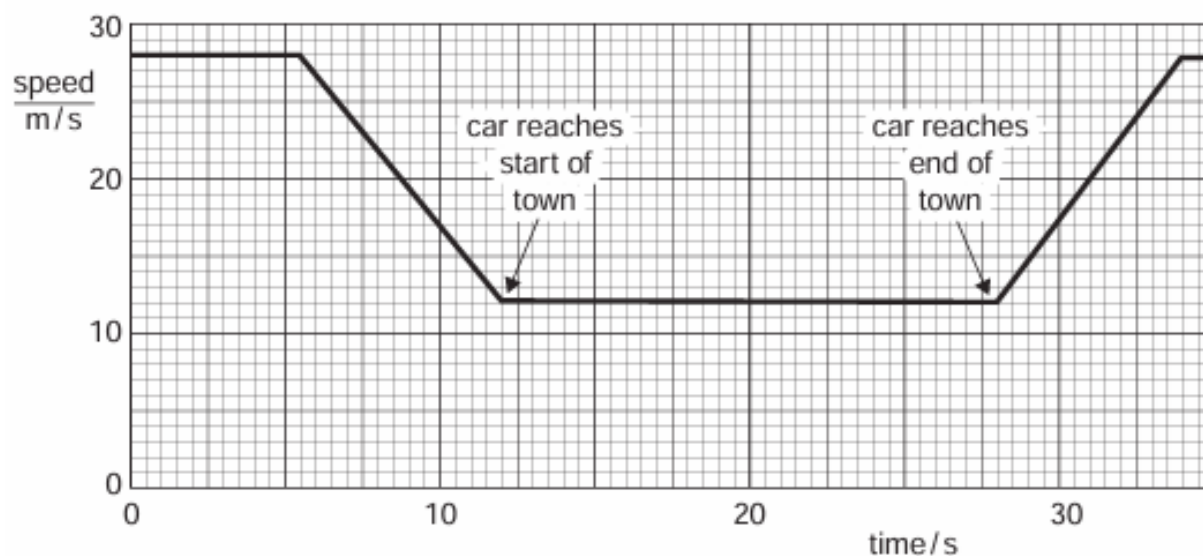


Fig. 1.1

- (a) (i) Determine the speed of the car in the town.

speed = [1]

- (ii) Determine the time taken by the car to pass through the town.

time taken = [1]

- (iii) Calculate the distance travelled by the car in the town.

distance = [1]

- (b) The car accelerates after passing through the town.

Calculate the acceleration. Give the unit of your answer.

acceleration = [3]

35. M/J/06/Q2

A cyclist starts from rest. He accelerates and then travels at a constant speed. At 12 s, the cyclist applies the brakes and slows down. Photographs are taken of the cyclist at 4 s intervals. Fig. 2.1 shows the results.

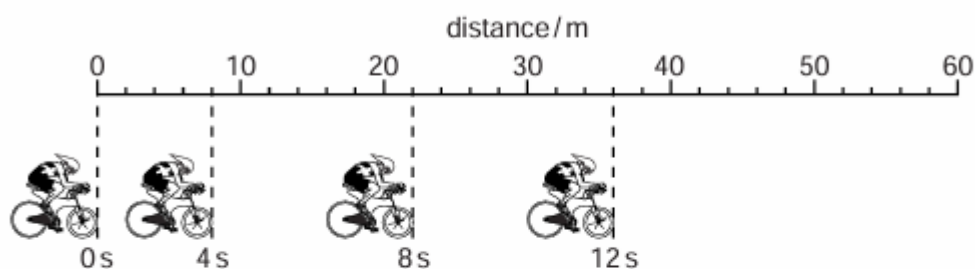


Fig. 2.1

- (a) On Fig. 2.1, draw a possible position of the front wheel of the cycle at 16 s. [1]
- (b) On Fig. 2.2, plot a distance-time graph of the cyclist for the first 16 s. [4]

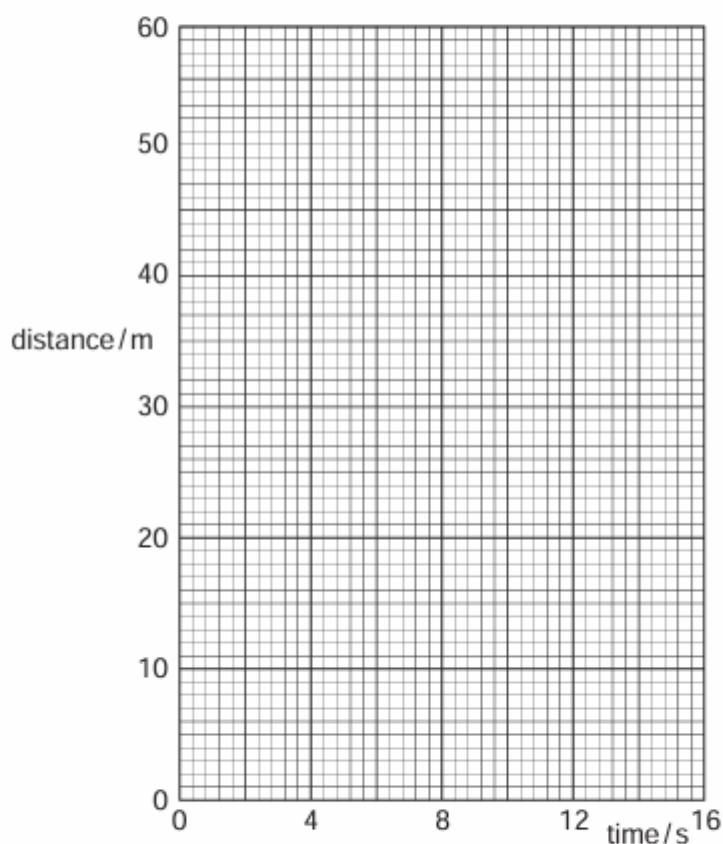


Fig. 2.2

- (c) Calculate the average speed of the cyclist during the first 12 s.

average speed = [2]

36. O/N/04/Q1

Fig. 1.1 shows a simplified speed-time graph for a train that travels between two stations.

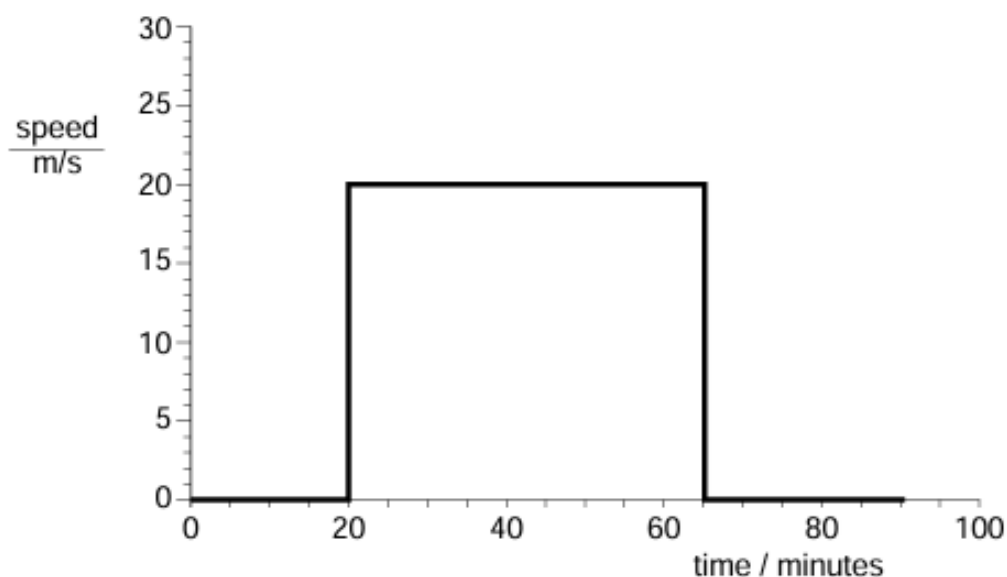


Fig. 1.1

- (a)** Describe the motion of the train.

.....

.....

.....

.....

.....[2]

- (b)** Calculate the distance travelled by the train between the two stations.

distance = [2]

- (c)** Another train travels between the same two stations on a parallel track. This train travels at a constant speed. It starts its journey at time $t = 0$ and finishes at $t = 90$ minutes.

On Fig. 1.1, draw the speed-time graph for this train. [2]

37. M/J/04/Q4

Fig. 4.1 shows a student clapping in front of a vertical wall. The wall reflects the sound.

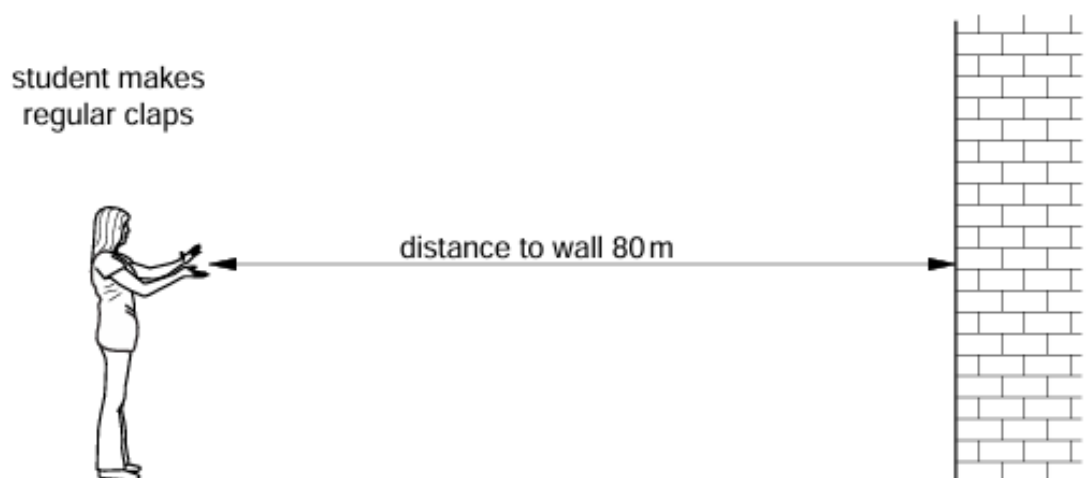


Fig. 4.1

The student changes the number of claps made in 1 minute until the reflection of each clap returns to her at exactly the same time as she makes the next clap.

The speed of sound in air is 330 m/s.

(a) Explain what is meant by *speed*.

.....
.....[1]

(b) Calculate the time between claps.

time = [3]

(c) Calculate the number of claps in 1 minute.

number of claps = [2]

ANSWERS

1.

Question	Answer	Marks
1(a)	uniform / constant acceleration (for 6 s) / acceleration of 3.7 m/s^2	B1
	after 6(0 s) constant speed / zero acceleration	B1
1(b)	$(a =) (v-u) / t$ in any form numerical or algebraic or $3.7 \text{ (m/s}^2\text{)}$ seen	C1
	$(m =) F / a$ or $(m =) Ft / v$ in any form numerical or algebraic	C1
	490 (kg)	A1
1(c)	area under graph or average speed $\times$ time numerical or algebraic	C1
	66 (m)	A1

2.

Question	Answer	Marks
1(a)	speed is constant for 10 s / at first	B1
	deceleration - has (average) value 0.39 m/s^2 - lasts for 40 s - is non-uniform	B1
1(b)(i)	friction (with road) and <u>air</u> resistance or (air) drag	B1
1(b)(ii)	resistive force(s) change / decrease (as speed decreases) or (graph shows) deceleration is not constant / decreases or non-uniform decrease in speed	B1

3.

Question	Answer	Marks
7(a)	acceleration / increase in speed and gravity / weight mentioned	B1
	air resistance / friction increases (with speed)	B1
	terminal / constant velocity reached or air resistance (becomes) equal to accelerating force	B1
7(b)	(distance travelled =) area under the graph or $(x =) \frac{1}{2}(u + v)t$ or $\frac{1}{2} \times 15.2 \times 4.0$	C1
	$35 - (\frac{1}{2} \times 15.2 \times 4.0)$ or 30.4 or 30	C1
	4.6 m	A1
7(c)(i)	(magnitude of deceleration =) gradient or Dv / Dt or $15.2 / 4.0$ or equivalent numbers (from graph) in same ratio	C1
	$15.2 / 4.0$ or equivalent numbers in same ratio	C1
	3.8 m/s^2	A1
7(c)(ii)	$(F =) ma$ or $(46 + 9.0) \times 3.8$ or 55×3.8	C1
	210 N	A1
7(c)(iii)	<u>from</u> kinetic energy (and from gravitational potential energy)	B1
	<u>to</u> thermal energy and no intermediate energy	B1
7(d)	due to his mass or mass mentioned	B1
	(his) mass resists a change to its state of motion	B1

4.

Question	Answer	Marks
1(a)	(distance =) area (under graph) or 77×56 or $\frac{1}{2} \times 77 \times 70$ or 4300 or 2700 or $\frac{1}{2} \times 77 \times (56 + 126)$	C1
	$77 \times 56 + (\frac{1}{2} \times 77 \times 70)$ or $4300 + 2700$ or $\frac{1}{2} \times 77 \times (56 + 126)$	C1
	7000 m or 7.0×10^3 m or 7.0 km	A1
1(b)(i)	(a =) $\Delta v / \Delta t$ or $77 / 70$ (etc.)	C1
	1.1 m / s ²	A1
1(b)(ii)	(F =) ma or $3.8 \times 10^5 \times 1.1$	C1
	4.2×10^5 N	A1

5.

Question	Answer	Marks
7(a)(ii)	area under curve or formula for area of a triangle / trapezium seen or working shown used	B1
	(accelerating) 80 (m) or (at constant speed) 160 (m) or (decelerating) 32 (m) seen	C1
	(total distance) 270 or 272 m	A1
7(a)(iii)	(average speed) = <u>total</u> distance / (total) time algebraic or numerical	C1
	11 m / s	A1
7(a)(iv)	horizontal line drawn from 0 to 24 s at any level or horizontal line at the value calculated in 7(a)(iii) for at least 16 s	B1
	horizontal line from 0 to 24 s at the value calculated in 7(a)(iii)	B1

6.

Question	Answer	Marks
1(a)	scale vector diagram showing 12 N, 36 N and correct resultant force in correct direction	B1
	38 ± 1 kN	B1
	$018 \pm 3^\circ$	B1
1(b)(i)	(only in uniform acceleration) same <u>increase</u> / <u>change</u> in speed / velocity or has a constant acceleration or <u>speed-time graph</u> is a straight line	B1
	(only in uniform acceleration) same <u>increase</u> / <u>change</u> in speed / velocity in <u>same time</u> (period) or same rate of change of speed / velocity (with time) or <u>change in</u> speed / velocity is proportional to <u>time</u>	B1
1(b)(ii)	(a =) F / m in any way, algebraic or numerical	C1
	0.63–0.66 m / s ²	A1

7.

Question	Answer	Marks
2(a)	1.4 s	B1
2(b)(i)	same change in speed / velocity in the same time	B1
2(b)(ii)	slope / gradient changes	B1
2(c)	any clear statement or any use of area under graph or $d = s \times t$ or an (average) speed $\times$ time	C1
	calculation of any area between 1.4 and 7.0 s on Fig.2.1 e.g. 14×2.6 ; 36.4; 4×3 ; 12; $\frac{1}{2} \times 12 \times 2.6$; 15.6; 2.6×8 ; 20.8 (m) seen	C1
	48 m	A1

8.

Question	Answer	Marks
1(a)(i)	accelerating or increasing speed	B1
	decreasing acceleration or speed increasing at a decreasing rate or reaches a constant speed	B1
1(a)(ii)	<u>initially</u> acceleration is due to force of gravity / weight (only) or force of gravity / weight larger than air resistance from ($t = 0$ to ≈ 7 s)	B1
	air resistance increases (with speed)	B1
	resultant force decreases / becomes zero or forces balance or air resistance equals weight	B1
1(b)	two different pairs of co-ordinates from <u>straight-line</u> section of graph seen	C1
	correct use of co-ordinates in a division	C1
	48 – 52 m / s	A1

9.

Question	Answer	Marks
7(a)	(between $t = 0$ and $t = 10$ s) it is accelerating or speed / velocity increasing	B1
	between $t = 35$ s and $t = 40$ s, it is decelerating or speed / velocity decreasing	B1
	magnitude of acceleration less than magnitude of deceleration or between $t = 0$ and $t = 10$ s starts from rest / zero speed or between $t = 35$ ms and $t = 40$ s it finishes at rest / with zero speed	B1
7(b)(i)	two pairs of co-ordinates from the straight-line section	C1
	appropriate division using candidate's coordinates	C1
	$11.7 \leq \text{speed} \leq 12.0$ m / s	A1
7(b)(ii)	<u>total</u> distance / <u>total</u> time or (distance =) 390 (m) and 39 (s) $\leq$ (time =) $\leq$ 40 (s)	C1
	9.7 to 10.0 m / s	A1

10.

Question	Answer	Marks
1(a)	constant gradient	B1
1(b)	use of area (under graph) or $\frac{1}{2}vt$ or average speed 4.5 (m/s)	C1
	9(.0) m	A1
1(c)	line of decreasing slope after 2 s	B1
	horizontal line at 12 m/s from 5.5 to 8 s	B1
	line, curved or straight, from (8, 12) to (11,10)	B1

11.

Question	Answer	Marks
1(a)	weight is directly proportional to mass or $W = mg$	B1
	a larger force acts on the more massive body or $a = F / m$ or $a = W / m$	B1
	mass cancels or acceleration inversely proportional to mass	B1
1(b)(i)	$(\Delta v =) at$ or 1.6×1.5	C1
	2.4 m / s	A1
1(b)(ii)	2.4 marked on y-axis and straight line from (0, 0) to (1.5, 2.4)	B1
1(b)(iii)	$\frac{1}{2} \times 2.4 \times 1.5$ or area under the graph mentioned / attempted	C1
	1.8 m	A1

12.

Question	Answer	Marks
1(a)(i)	58 km	B1
1(a)(ii)	0.8(0) hr or 48 min	B1
1(b)	steep(er) / larg(er) slope / gradient or travels a larger distance in the same / unit time or takes a shorter time to travel the same distance or covers more distance in less time	B1
1(c)	$(s =) d / t$ numerical or algebraic; 58 / 4	C1
	15 km / hr	A1

13.

Question	Answer	Marks
9(a)(i)	10.2–10.4 (cm) or 50 (km) or 15 (km) seen	C1
	51–53 km	A1
	direction (0)72–74° or N 72–74 °E etc.	B1
9(a)(ii)	(distance) has no direction or (distance) is a scalar	B1
9(a)(iii)	changes direction or goes round a corner	B1
	velocity changes or velocity depends on direction or acceleration / force is towards the centre of circle / centripetal	B1

14.

- | | | |
|----------|---|-----------|
| (a) | points plotted correctly at (1,0.5) (2,1.6) (3,3) (4,5) and (5,7) | B1 |
| | smooth curve from origin | B1 |
| (b) (i) | straight line | B1 |
| | or gradient / slope constant | |
| | travels equal distances in same time | B1 |
| | or gradient equals speed | |
| (b) (ii) | weight and force (upwards) from oil / liquid | B1 |
| | forces balanced / cancel / are equal (and opposite) | B1 |
| | or no resultant (force) | |

15.

- (a) distance measured (between 2 marks) with metre rule or tape measure **B1**
time measured (between 2 marks) with stopwatch / stopclock / timer **B1**
(average speed =) (total) distance / time **B1**
- (b) (i) decrease in speed **B1**
or deceleration
- (b) (ii) $(\Delta v =) a \times t$ in any form numerical or algebraic **C1**
4.5 m / s **A1**

16.

Question	Answer	Marks
1(a)	$(F =) ma$ or 35×2.6	C1
	91 N	A1
1(b)(i)	straight line from origin to dashed line / t_1	B1
	gradually decreasing gradient until the line is horizontal	B1
1(b)(ii)	area <u>under</u> the line or area between line and x-axis	B1

17.

- (a) (i) $(a =) v(-u) / t$ or $25 / 14$ **C1**
1.8 m / s² **A1**
- (ii) initial straight line from (0,0) to (14,25) **B1**
gradient of line decreases after 14 s **B1**
and flat from (70,55) to (80,55)

18.

- (a) $(v = u +)at$ or 3.4×5.0 **C1**
17 m/s **A1**
- (b) (i) 0 or zero or no resultant force **B1**
- (ii) straight line of positive gradient from (0, 0) **B1**
horizontal line at $v > 0$ and after initial acceleration **B1**
straight line from (0, 0) to (5.0, 17) and **B1**
straight line from (5.0, 17) to at least (15.0, 17)
- (iii) calculate the area **under** the graph or area of trapezium **B1** [7]

19.

- (a) (i) $(a =) \Delta v / t$ or $95 / 0.011$ **C1**
 $8.6(3636) \times 10^3$ m/s² **A1**
- (ii) $(F =) ma$ or $0.018 \times 8.63 \times 10^3$ **C1**
150 / 155(.4545) / 160 N **A1**
- (b) line from (0, 0) to (0.011, 95) with decreasing gradient **B1**
(becomes) horizontal at (0.011, 95) **B1**

20.

- | | | |
|---------|---|----------|
| (a) (i) | 60 m | B1 |
| (ii) | 12 s | B1 |
| (b) (i) | straight line from origin to 200 m at 40 s
any line straight or curved from (40,200) to (60,500) | B1
B1 |
| (ii) | $s = d/t$ or 500/60
8.3 m/s | C1
A1 |

21.

- | | | |
|---------|--|----------------|
| (a) (i) | distance (travelled) per second or speed
distance (travelled) per second/speed in a given direction
or displacement/time
or change in displacement per unit time
or displacement (travelled/covered) per unit time
or rate of change of displacement | C1
A1 |
| (ii) | opposite direction | B1 |
| (iii) 1 | value seen for v and corresponding value of t
$0 < t \leq 1.4$ and $0 < v \leq 14$
($a=$) $v-u/t$ algebraic or numerical equation
10 m/s^2 | C1
C1
A1 |
| 2 | sensible comment | A1 |

22.

- | | | |
|---------|---|----------------|
| (a) | changing speed/velocity
change in speed/velocity/time constant or $(v-u)/t$ constant or constant/equal
rate of change of speed/velocity | C1
A1 |
| (b) | (a vector quantity has) direction | B1 |
| (c) (i) | 1. X between $t \geq 0$ and $t < 10$ s
2. Y between $t > 20$ s and $t < 30$ s
3. Z between $t > 10$ s and $t < 20$ s or between $t > 30$ s and $t < 40$ s | B1
B1
B1 |
| (ii) | 1. two speed values from graph between 15 and 35 s (± 1 mm)
two corresponding time values from graph between 15 and 35 s
(± 1 mm) or ($a =$) $\Delta v/t$
500 m/s^2 | C1
C1
A1 |

23.

- | | | |
|-----|---|------|
| (a) | straight line from (0, 0) to (3, 2.4) | [B1] |
| | horizontal line from 3 s to 8 s | [B1] |
| | straight line from end of a horizontal line to zero in 1 s | [B1] |
| (b) | constant/same increase in velocity or constant change in velocity | [C1] |
| | constant/same increase in velocity per sec/unit time | [A1] |
| (c) | occurs in a short(er) time | |
| | or acceleration took 3 s and deceleration took 1 s | [B1] |
| (d) | (d =) speed $\times$ time numerical or algebraic or area under graph clear | [C1] |
| | 1.2×3 or 3.6 (m) or 2.4×5 or 12 seen | [C1] |
| | 15.6 m or 16 m | [A1] |
| (e) | (i) mgh seen in any algebraic or numerical form, e.g. $30 \times 10 \times 1.6$ | [C1] |
| | 480 J | [A1] |
| | (ii) heat or thermal energy or sound produced | |
| | or work done against friction / air resistance | [B1] |
| (f) | at least two distances and corresponding times mentioned | [C1] |
| | how the actual measurement is made, e.g. (any one from) | |
| | • make mark on ground every second and measure distances | |
| | • note video position every sec and use a scale to find distances | |
| | • make mark on ground every meter and measure/take time as girl passes | [A1] |
| | how constant speed is proved using measurement, e.g. (any one from) | |
| | • same distance between each position for the same time interval | |
| | • same time interval for equal distances | |
| | • $\Delta d / \Delta t$ constant or slope of distance-time graph constant | [B1] |

24.

- | | | |
|-----|---|----|
| (a) | (i) start at origin and not horizontal | B1 |
| | gradient (gradually) decreasing (ignore sudden decrease) | |
| | (not if part of curve above horizontal section) | B1 |
| | final horizontal section (≥ 1 cm) (not if v is shown as $\neq 40$ m/s) | B1 |
| | (ii) area under the graph or count squares under graph | M1 |
| | between $t = 0$ and horizontal section or when speed is changing or | |
| | calculate equivalent distance to 1 cm^2 (after counting squares) | A1 |

25.	
(a) travels further in each second / in same time / between images	B1
(b) (s=) d/t in any form algebraic or numerical 40 cm/s; 0.4(0) m/s	C1 A1
26.	
(a) 72 m/s	B1
(b) (i) area (under graph) or $\frac{1}{2}$ base $\times$ height or $\frac{1}{2}vt$ or $\frac{1}{2} \times 9 \times 72$ 320/324 m	C1 A1
(ii) change in velocity/time or $\Delta v/t$ or 72/9 8(.0) m/s ²	C1 A1
27.	
(a) (i) curve with decreasing gradient from origin to 50 m/s at 10 s	B1
constant speed from 10 to 20 s	B1
decrease to 5 m/s at 25 s	B1
constant speed from 25 s until at least 30 s	B1
(ii) gradient/slope not constant/decreases // graph curves // graph not a (straight) line // increase (in speed) per second/unit time not equal	B1
(b) any mention of air resistance/drag/upward force	B1
(initially) force upwards larger than force downwards // resultant force upwards	B1
air resistance decreases (with fall in speed)	B1
(at constant speed) air resistance/friction/drag equals weight //	
forces (up and down) balance // zero resultant force	B1
(c) 500 m	B1
(d) (i) $(a =) \frac{v - u}{t}$ in any numerical or algebraic form, e.g. 45/5	C1
9(.0) m/s ² ecf (a)(i)	A1
(ii) $(F = ma)$ in any numerical or algebraic form, e.g. 60×9 ecf (i)	C1
540 N	A1
(iii) area under graph/line/curve	B1
28.	
(a) (i) 11.5 m/s	B1
(ii) decrease in speed // negative gradient	C1
equal changes/decreases in speed in the same time // const. neg. grad. on v-t graph	A1
(b) (i) flat line at 18 m/s from $t = 0$ to 15	B1
constant slope downwards parallel to initial line (by eye)	B1
(ii) greater area under graph // higher initial/average speed	B1

29.

- (a) velocity has a direction/is a vector **or** speed does not have a direction/is not a vector
or displacement/time **and** distance/time
 (ign speed is a scalar) B1
- (b) (i) (–) 47 m/s B1
- (ii) ($a =$) v/t **or** $47/0.0013$ C1
 (–) $3.6(1538 \text{ etc.}) \times 10^4 \text{ m/s}^2$ A1

30.

- (a) (i) any **one** time between 1.60 and 2.50 s **or** range of correct values B1
- (ii) any **one** time between 0.75 and 1.65 s **or** range of correct values B1
- (iii) 2.5(0) s B1
- (b) area (under graph) **or** $\frac{1}{2}bh$ **or** $\frac{1}{2}gt^2$ **or** $\frac{1}{2} \times 0.75 \times (7.3 \text{ to } 7.5)$ C1
 2.7(375) to 2.8(125) m A1

31.

- (a) (speed) increases or (paper) accelerates
 (speed) becomes constant/uniform or acceleration zero (after 0.5 s) B1
- (b) any clear change in distance/time or 1.87 (m/s) (allow 1.9) B1
 2.3–2.5 m/s C1

32.

- (a) parachute opens or speed drops from (50 to 5 m/s) **or** decelerates (**e.g.** uniformly) **and**
 lands/hits ground **or** speed becomes 0 or stops (**e.g.** decelerates) B1
- (b) accelerates **or** speed increases (**not** increasing acceleration) B1
 acceleration decreases (to 0) **or** speed becomes constant B1
- (c) forces balance/cancel **or** no resultant **or** equal and opposite (**not just** forces equal) B1
 weight/gravity **and** air resistance/drag mentioned (**not** upthrust/friction) B1
- (d) ($d =$) st **or** $s=d/t$ or any speed $\times$ any time **or** area under graph C1
 150 m A1

33.

- (a) accelerates or speed increases **from rest/for 2-4s/for 8-20m** B1
 then a constant/uniform speed **or** velocity B1
- (b) between 7 and 8 m B1
- (c) distance 80 (+ 2) or $s = d/t$ in any algebraic or numerical form C1
 7.3 **or** 7.27 **or** 7.273 m/s A1

34.		
(a)	(i) 12 m/s	B1
	(ii) 16 s	B1
	(iii) 192 m or (i) × (ii)	B1
(b)	a = (v-u)/t in any format e.g. numerical (allow 4 clearly attributable wrong numbers) or gradient of v-t/the graph	C1
	2.7 (2/3 sig. fig. only, do not accept fraction, cao)	A1
	m/s ²	B1
35.		
	any position before 50 m	B1
(b)	points plotted correctly at 4,8 and 12 s (± ½ square)	B1
	from origin to 4 sec curve drawn	B1
	from 4 to 12 sec straight line positive gradient	B1
	from 12 to 16 sec gradient decreases (but not –ve)	B1
(c)	speed/time	C1
	3 m/s	A1
36.		
(a)	speed uniform or 20 m/s	B1
	stationary/not moving till 20 minutes or after 65 minutes or moves for 45 minutes	B1
	(not if inconsistent; all times ±2 min; ignore acceleration/deceleration periods)	
(b)	d = st any algebraic or area calculated	
	or 20 x 45, 20 x 90, 20 x 45 x 60, 20 x 90 x 60	C1
	54 000 m or 54 km	A1
(c)	any constant speed from 0 to 90 minutes (may stop at 90 or go down to axis)	M1
	10 m/s (no e.c.f. b)	A1
37.		
(a)	distance traveled per unit time or in one second / distance ÷ time	B1
	or rate of change of distance	C1
(b)	s = d/t in any algebraic or numerical form	C1
	any doubling of distance or final time	A1
	0.48 s (allow 0.24s 2/3 accept 0.5s)	C1
(c)	60/0.48 (5)	C1
	123.75 accept 120, 123, 124 (ecf (b))	A1